Attention-Aware Task Scheduling via Deep Reinforcement Learning under Partial Observability for Human-Centric Productivity Optimization

P. N. Bhargav Teja[a], Lanka Sree Chathurya[a], Rajay Vedaraj I.S.[a,*]

[a]*Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

Abstract

Dispatching rules such as earliest-deadline-first (EDF) and shortest-processing-time (SPT) assume constant processing capacity, which fails when the resource is a person whose attention depletes, whose fatigue accumulates and whose alertness follows a circadian rhythm. We formulate human-centric task scheduling as a partially observable Markov decision process in which rest is an explicit action, attention and fatigue are latent states observed only through noisy estimates, and the reward balances deadline-weighted task value against cognitive load. We propose an attention-aware deep reinforcement learning framework that approximates the belief state with an observation history, and instantiate it with a value-based learner (AA-DQN) and a policy-gradient learner (AA-PPO). In a literature-grounded simulator, both were compared with nine rule-based schedulers, including grid-tuned EDF and SPT rules with periodic or state-triggered rest, using ten training seeds, 500 held-out workdays with common random numbers, and multiplicity-corrected tests on reward-independent metrics. Rules that never rest performed poorly (EDF: 46.8% of tasks on time). AA-DQN matched the best tuned rule in return and completed more tasks on time. AA-PPO exceeded the best rule in return (+0.73, $p < 10^{-4}$), on-time completion (75.1% versus 71.4%) and difficulty-weighted

*Corresponding author.
Email address: rajay@vit.ac.in (Rajay Vedaraj I.S.)

on-time completion (+7.0 percentage points). The learned policies alone combined high throughput with protection of demanding tasks, by doing hard work while attention is high and resting before and during the post-lunch dip. AA-PPO remained the best method under sensor noise, different worker profiles, heavier workloads and ±30% changes of every cognitive parameter, whereas AA-DQN generalised poorly to unseen workloads.

Keywords: Human-centric scheduling, Deep reinforcement learning, Partially observable Markov decision process, Cognitive fatigue, Attention, Rest-break scheduling, Industry 5.0

Nomenclature

Abbreviations

	AA-DQN	attention-aware deep Q-network	MDP	Markov decision process
	CI	confidence interval	POMDP	partially observable MDP
	CRN	common random numbers	PPO	proximal policy optimisation
	DQN	deep Q-network	Adam	adaptive moment estimation
	DRL	deep reinforcement learning	RL	reinforcement learning
	EDF	earliest deadline first	SPT	shortest processing time
11	FIFO	first in, first out	TD	temporal difference
	GRU	gated recurrent unit	HRC	human–robot collaboration
	AA-PPO	attention-aware PPO	GAE	generalised advantage estimation
	OT	on-time rate	WOT	difficulty-weighted on-time rate
	CR	completion rate	HFE	high-fatigue exposure
	BS	break share	pp	percentage points

12 **Symbols**

Workload and time

t, T, Δ	decision slot index, slots per day ($T = 48$), slot length ($\Delta = 10$ min)
h_t	clock time of slot t in hours, $h_t = 9 + t\Delta/60$
N	number of tasks per day
e_i, d_i, w_i, D_i	arrival (entry) slot, difficulty, work content and deadline of task i
s_i	relative deadline slack of task i
$R_{i,t}, \tau_i$	remaining work and completion slot of task i
$\mathcal{P}_t, \mathcal{E}, \mathcal{H}$	available unfinished tasks; easy tasks; hard tasks
$C_t, \mathcal{M}_t$	tasks completed / deadlines missed in slot t

Cognitive state and observation

H_t	homeostatic attention resource (latent)
$A_t, c(h)$	effective attention (latent); circadian modulation
F_t	fatigue (latent)
ρ_t, ρ_0	work rate (work units per slot); work rate of a fully rested worker
α, ϕ, β	attention depletion rate, fatigue coupling, attention recovery rate
γ	circadian amplitude
δ, η, κ	fatigue accumulation rate, fatigue recovery rate, fatigue throughput penalty
σ_p, ξ_t	process-noise scale and standard normal process noise
σ_o, ϵ_t	observation-noise scale and standard normal observation noise
$\tilde{A}_t, \tilde{F}_t$	noisy observations of attention and fatigue
u_t, b_t	indicators of working and resting in slot t
o_t, x_t, k	observation vector, observation history, history length
$n_t, W_t, \sigma_t, m_t, \omega_t$	per-class queue length, remaining work, normalised slack, overdue count, shortest remaining work

Decision problem and learning

$\mathcal{S}, \mathcal{A}, \Omega$	state, action and observation spaces
P, O, R	transition, observation and reward functions
$a_t, g(\cdot)$	macro-action and dispatch function

1. Introduction

Scheduling decides which piece of work is executed when, and it is one of the oldest problems in operations research and computer engineering [1–4]. Classical dispatching rules such as First-In-First-Out (FIFO), Shortest Processing Time (SPT) and Earliest Deadline First (EDF) are attractive because they are transparent, cheap to compute and, under the right assumptions, provably optimal: EDF minimises maximum lateness on a single resource with constant capacity [2, 5], and SPT minimises mean flow time [1]. The common assumption underlying these guarantees is that the resource executing the work has a *fixed, state-independent processing capacity*. This assumption is reasonable for processors and machines but is violated whenever the resource is a person.

Human work capacity is dynamic. Sustained effort depletes attentional resources and produces the vigilance decrement first documented by Mackworth [6] and since replicated across many tasks [7–9]; mental fatigue accumulates with time-on-task and degrades both speed and accuracy [10–13]; and alertness follows a circadian rhythm with a pronounced early-afternoon dip [14–17]. Rest breaks partially reverse these effects, and field studies consistently report that well-timed breaks preserve or even increase productivity despite the time they consume [18–22]. A schedule that is optimal for a constant-capacity machine can therefore be poor for a human: it front-loads demanding work regardless of cognitive state, never allocates recovery, and drives the worker into a low-capacity regime in which deadlines are missed precisely because capacity was ignored. The concern is not only productivity. Chronic fatigue is associated with errors and accidents [23, 24], and the move from Industry 4.0 towards a human-centric Industry 5.0 explicitly calls for production and service systems that adapt to the well-being of

their workers rather than the reverse [25–29].

Operations-management research has long recognised this gap. Models of worker fatigue and recovery have been integrated into dual-resource-constrained systems [30–32], rest-allowance and break-scheduling problems have been formulated analytically [33–35], and taxonomies for merging scheduling theory with human factors have been proposed [36–39]. These approaches, however, usually assume that the worker's state is known or deterministic and optimise an offline plan; they do not provide an adaptive *policy* that reacts to the evolving, uncertain state of a specific person during the day. Deep reinforcement learning (DRL), in turn, has produced effective adaptive schedulers for job shops, clusters and production lines [40–46], and recent surveys document rapid growth of the field [47–49]. Yet the resources in these studies are machines, and the few recent studies that include worker fatigue embed it in large shop-floor models where fatigue is a deterministic, fully observed variable [50, 51]. Two features of the human problem are thus rarely addressed together: *the decision of when to rest is itself a scheduling action*, and *the cognitive state that should drive that decision cannot be measured directly*. In practice, attention and fatigue are inferred from noisy proxies such as physiological signals, eye tracking, performance probes or self-report [52–56], which makes the scheduling problem partially observable.

This paper studies attention-aware task scheduling for a single knowledge or production worker as a partially observable Markov decision process (POMDP) [57–59] and solves it with a memory-augmented DRL framework that we instantiate with a value-based learner, an attention-aware deep Q-network (AA-DQN), and a policy-gradient learner, attention-aware proximal policy optimisation (AA-PPO). We pay particular attention to the methodological issues that make DRL

comparisons hard to trust [60, 61]: we compare against strong, *tuned*, cognition-aware heuristics rather than only against FIFO and EDF; we use disjoint seed sets for training, tuning and testing; we evaluate on 500 held-out workdays with common random numbers; we report ten independent training seeds with confidence intervals and multiplicity-corrected paired tests; and we separate effects of the reward function from effects of the dynamics through ablation.

The contributions are as follows.

(i) **A cognitively grounded scheduling simulator.** We formulate a workday-scale scheduling environment in which a latent homeostatic attention resource, an exponentially accumulating and recovering fatigue state [30], and a circadian process [14, 15] jointly determine the worker's instantaneous work rate. Tasks arrive stochastically with heterogeneous difficulty, work content and deadlines. Rest is an explicit action. The simulator, its parameters and their behavioural rationale are fully specified and released.

(ii) **A POMDP formulation with a multi-objective reward.** The scheduler observes only noisy estimates of attention and fatigue together with a compact, size-invariant summary of the task queue. The reward combines deadline-weighted task value with explicit attention and fatigue terms, and we analyse the resulting productivity–fatigue trade-off as a Pareto front [62, 63].

(iii) **An attention-aware DRL framework with two learners.** A finite observation history acts as an approximate sufficient statistic for the belief state [64–66]. AA-DQN combines it with Double Q-learning [67] and a dueling architecture [68]; AA-PPO combines it with a clipped policy-gradient learner [69]. A component ablation quantifies the contribution of each

element.

(iv) **A rigorous empirical study.** Against nine rule-based baselines, including grid-tuned heuristics that combine EDF or SPT sequencing with periodic or state-triggered rest and observe the same noisy cognitive signals, and across the two learners, we test (a) whether learning helps, (b) *why* it helps, through environment ablations that remove observation noise, the circadian process, the cognitive dynamics or the cognitive reward terms, (c) what behaviour is learned, and (d) how the learned policy generalises to unseen worker profiles, workloads and sensor-noise levels, and (e) how sensitive the results are to the agent's hyperparameters and to every parameter of the cognitive model.

Novelty with respect to existing approaches. Fatigue has previously been included in scheduling models, so it is worth stating precisely what is new here. (1) *Rest as a learned scheduling decision.* In fatigue-aware shop scheduling and dual-resource models [30, 50, 51, 70, 71], fatigue is a constraint or a cost attached to a given assignment, and recovery happens in predefined breaks. In our formulation the scheduler decides *when* to rest, and recovery is endogenous. (2) *Partially observable cognitive state.* Existing fatigue-aware schedulers and many human–robot task-allocation methods compute fatigue from the schedule or assume it is measured exactly [50, 51, 72]. We treat attention and fatigue as latent, model the estimation error explicitly and show that memory is needed to act on noisy estimates. (3) *Attention, fatigue and circadian rhythm jointly determine capacity.* Previous models typically include fatigue alone. Our productivity model makes demanding tasks disproportionately sensitive to low attention, which creates the difficulty–capacity matching problem that the learned policy solves. (4) *Evaluation*

against tuned, cognition-aware rules. DRL schedulers are usually compared with static dispatching rules [47, 49]. We compare with nine rules, including grid-tuned rules that rest and that observe the same signals as the agent, and we instantiate the framework with two DRL algorithms. We also report where learning does not help and how sensitive the conclusions are to the simulator parameters. To our knowledge, no previous study combines these four elements (Table 1).

The remainder of the paper is organised as follows. Section 2 reviews related work and positions the study. Section 3 formulates the problem. Section 4 presents the framework and its two learners. Section 5 details the experimental protocol. Section 6 reports results, Section 7 discusses implications, ethics and limitations, and Section 8 concludes.

2. Related work

2.1. Classical and dynamic scheduling

Single-resource sequencing theory provides the canonical dispatching rules and their optimality conditions: SPT for mean flow time [1], EDF for maximum lateness [2], and the Moore–Hodgson algorithm for the number of late jobs [73]. The three-field classification of Graham et al. [3] and the textbook treatment of Pinedo [4] organise the deterministic theory. In real-time computing, EDF is optimal for preemptive scheduling of independent tasks whenever the processor is not overloaded [5], but its performance can collapse under overload, when a domino effect causes many deadlines to be missed [74]. This failure mode is directly relevant here, because a fatigued worker is in effect an overloaded resource. Surveys of priority rules [75, 76] and of dynamic and reactive scheduling [77, 78] show that practice relies heavily on such rules when information arrives online. In

all of these models, however, processing times are exogenous: they do not depend on the history of the schedule.

2.2. Human factors in scheduling and workforce models

Personnel scheduling and rostering [79–81] assign workers to shifts but typically treat performance within a shift as constant. Research at the interface of operations and human resources argues that this abstraction misses first-order effects [37, 38], and Lodree et al. [36] proposed a taxonomy for integrating human factors into scheduling theory. Quantitative models of learning, forgetting, fatigue and recovery have been developed for dual-resource-constrained systems [30–32, 82]. They use exponential fatigue accumulation during work and exponential recovery during rest, which is the functional form we adopt. Rest-break scheduling has been formulated as an optimisation problem since Bechtold et al. [33], with ergonomic guidance on work–rest ratios summarised by Konz [34] and rest-allowance models for task assignment developed by Calzavara et al. [35]. Fatigue has also been incorporated into flexible job-shop formulations solved by metaheuristics [50], by evolutionary algorithms with learned neighbourhood operators that account for worker transportation fatigue [70], by graph-based DRL [51], and into hybrid flow shops solved with an offline DQN-based metaheuristic that trades fatigue against energy [71]. Related work schedules flow shops with human factors and competing agents using DRL [83], handles multi-skilled worker constraints in dynamic flexible job shops [84], and integrates human feedback or preferences into RL-based scheduling [85, 86]. In human–robot collaborative assembly, workstation schedules under ergonomic fatigue constraints [72] and RL-based task allocation using digital twins, human feedback, attention-based hierarchical policies or safe RL with fatigue prediction [87–90] have recently been proposed. Reviews of hu-

man factors in Industry 4.0 and in order picking [39, 91–93] conclude that worker heterogeneity and well-being are still under-represented in operational models. A common feature of this literature is that the worker state is assumed to be known, either deterministic or computed from the schedule, so the decision problem is fully observable.

2.3. Deep reinforcement learning for scheduling

Reinforcement learning has been applied to scheduling since the mid-1990s [94, 95], and the combination of Q-learning [96] with deep function approximation [97] enabled scheduling policies that operate on rich state representations. Notable examples include cluster resource management [40], data-processing job scheduling with graph neural networks [41], semiconductor production scheduling [42], dynamic flexible job shops with job insertions [43, 46], and size-agnostic job-shop dispatching [44, 45, 98]. Parameter-adaptive RL for dynamic job shops [99] and RL for combinatorial optimisation more broadly [100, 101] complete the picture. Offline RL has recently been applied to dispatching for job shops [102]. Systematic reviews [47–49] note three recurring weaknesses: comparisons against untuned or weak baselines, limited statistical analysis, and little attention to human resources. Our study addresses all three.

2.4. Cognitive state, its measurement, and adaptive systems

Resource theories of attention [103, 104] and cognitive load theory [105] model mental capacity as a limited resource. Vigilance research shows that this resource is depleted by sustained effort [6, 7] and partially restored by brief breaks [106]. Accounts of fatigue as an effort-regulation or opportunity-cost signal [12, 107] and evidence linking fatigue to disengagement [108] motivate treating fatigue as a

state variable that shapes future capacity. Biomathematical fatigue models, from the two-process model of sleep regulation [109–111] to the three-process model of alertness [14] and applied models such as SAFTE [112], combine homeostatic and circadian components. We borrow this decomposition at the time scale of a single workday. Recovery research in organisational psychology confirms the value of micro-breaks [22, 113–115]. On the measurement side, workload and fatigue can be estimated from EEG, cardiac, ocular and behavioural signals [52–54, 116, 117], from vigilance probes such as the PVT [55, 118], or from questionnaires such as NASA-TLX [56, 119]. All of these estimates are noisy. Adaptive automation and physiological computing use such estimates to adjust system behaviour [120–125], and interruption-management systems decide when to notify a user based on inferred cognitive state [126–129]. Reinforcement learning has been used to personalise just-in-time health interventions [130–133], and in human–robot collaboration to adapt task allocation to human capability and fatigue [134–138]. These works show that cognitive-state estimates can drive decisions, but they rarely address the sequential scheduling of a task queue under deadlines.

2.5. Partial observability and multiple objectives in RL

When the state is not directly observed, the optimal policy is a function of the belief state [57–59]. Memoryless policies can be arbitrarily poor [64]. Practical deep RL approximates the belief either with recurrent networks [65, 139, 140] or with finite observation histories, as in the frame stacking of Mnih et al. [97]. Ni et al. [66] show that well-tuned memory-based model-free agents are strong POMDP baselines. Stabilisation techniques for DQN, namely experience replay [141], target networks [97], Double Q-learning [67], dueling heads [68] and their combinations [142], are well established, as is the theory of TD learning with func-

tion approximation [143]. Scheduling for humans is inherently multi-objective: productivity, deadlines and well-being conflict. Population-based alternatives, such as large-scale multi-objective evolutionary algorithms with learned operators [144], approximate Pareto sets offline but do not provide a reactive policy for a single unfolding day. Multi-objective RL provides the vocabulary of scalarisation and Pareto fronts [62, 63, 145]. Reward shaping must be used with care, because additional reward terms can change the optimal policy [146]. Finally, the deployment of RL in settings that involve people raises safety and reliability concerns [147–149], which we address in Section 7.

2.6. Research gap

Table 1 positions this work against representative streams. To our knowledge, no prior study simultaneously (i) models dynamic attention, fatigue and circadian variation as endogenous state, (ii) treats rest as a scheduling action, (iii) assumes that the cognitive state is only partially observable, (iv) learns an adaptive policy, and (v) benchmarks that policy against tuned, cognition-aware heuristics with multi-seed statistical testing. Our study fills this gap.

Table 1: Positioning of this study relative to representative research streams (✓: addressed; ∘: partially or in a simplified form; –: not addressed).

Stream (representative works)	Dynamic human state	Rest as action	Partial observ.	Adaptive policy	Deadlines/ queue	Tuned cog. baselines
Dispatching rules [1, 2, 5, 76]	–	–	–	–	✓	–
Fatigue/recovery models in DRC systems [30–32]	✓	∘	–	–	∘	–
Rest-break / rest-allowance optimisation [33–35]	∘	✓	–	–	–	–
DRL for machine scheduling [40, 42–44, 46]	–	–	–	✓	✓	–
Fatigue-aware shop scheduling [50, 51, 70, 71]	✓	–	–	∘	✓	–
Fatigue-aware HRC task allocation [72, 87, 90]	✓	∘	∘	✓	∘	–
Adaptive automation / interruption mgmt. [120, 125–127]	✓	∘	✓	∘	–	–
This study	✓	✓	✓	✓	✓	✓

3. Problem formulation

We consider a single worker who processes a stream of tasks during one workday. A scheduler, which could be a digital assistant, a workflow-management system or a supervisory controller in a human–robot cell, decides at regular intervals what the worker should do next: start or continue an easy task, a hard task, or take a short break. The worker's capacity depends on a latent cognitive state that evolves with the schedule. All symbols and abbreviations are listed in the

Nomenclature, and Table 2 lists the parameter values.

3.1. Workload model

The day is divided into $T = 48$ slots of $\Delta = 10$ minutes (09:00–17:00). Each day contains $N = 12$ tasks. Six tasks are available at the start of the day ($e_i = 0$) and the remaining six arrive at slots e_i drawn uniformly from $\{1, \dots, \lfloor 0.6T \rfloor\}$, which models interruptions and new requests. Task i has difficulty $d_i \sim \mathcal{U}[0.2, 1.0]$ and is labelled *hard* if $d_i \geq 0.6$ and *easy* otherwise. Its work content and deadline are

$$w_i = 1 + 2.5\, d_i, \qquad D_i = \min\{T,\ e_i + \lceil s_i T \rceil\}, \qquad s_i \sim \mathcal{U}[0.3, 1.0], \tag{1}$$

so a task requires between 1.5 and 3.5 slots (15–35 min) at full capacity and its relative slack s_i ranges from 30% to 100% of the day. With these settings, the expected daily work content (≈ 30 units) is below the capacity of a fully rested worker (48 units) but above what a worker can deliver once attention and fatigue degrade. Scheduling decisions therefore matter: the day is *conditionally overloaded*.

3.2. Cognitive dynamics

Effective attention. Following the decomposition of alertness into homeostatic and circadian components [14, 109, 111], effective attention is

$$A_t = \mathrm{clip}\big(H_t + \gamma\, c(h_t),\ 0,\ 1\big), \qquad c(h) = 0.2\, e^{-(h-10.5)^2/(2\cdot 1.5^2)} - e^{-(h-14)^2/(2\cdot 1.2^2)}, \tag{2}$$

where $c(h)$ encodes a late-morning peak and the post-lunch dip around 14:00 [15–17] (Fig. 1a), and $\gamma = 0.15$ sets its amplitude.

Homeostatic attention resource. Working on a task of difficulty d depletes the attentional resource, and the depletion is accelerated by fatigue. This captures the vigilance decrement and resource-control accounts of sustained attention [6–8]. Resting restores the resource exponentially towards its maximum [22, 106]:

$$H_{t+1} = \mathrm{clip}\Big(H_t - \alpha\, d\, (1 + \phi F_t)\, u_t + \beta\, (1 - H_t)\, b_t + \sigma_p\, \xi_t,\ 0,\ 1\Big), \qquad \xi_t \sim \mathcal{N}(0, 1), \tag{3}$$

where $u_t, b_t \in \{0, 1\}$ indicate work and rest ($u_t + b_t = 1$), $\alpha = 0.02$ is the depletion rate per unit difficulty, $\phi = 0.5$ the fatigue coupling, $\beta = 0.30$ the per-slot recovery fraction, and $\sigma_p = 0.02$ the process-noise scale that represents unmodelled day-to-day and moment-to-moment variability.

Fatigue. Fatigue accumulates exponentially towards saturation during work and decays exponentially during rest. This is the discrete-time analogue of the fatigue–recovery model of Jaber and Neumann [30], also used in [31, 32]:

$$F_{t+1} = \mathrm{clip}\Big(F_t + \delta\, d\, (1 - F_t)\, u_t - \eta\, F_t\, b_t,\ 0,\ 1\Big), \tag{4}$$

with accumulation rate $\delta = 0.05$ and recovery rate $\eta = 0.20$.

Productivity. While working on task i, the worker completes

$$\rho_t = \rho_0\, A_t^{0.5 + d_i}\, (1 - \kappa F_t) \tag{5}$$

work units in slot t, with $\rho_0 = 1$ and $\kappa = 0.5$. The exponent makes hard tasks more sensitive to attention than easy ones (Supplementary Fig. S1), in line with resource theories in which demanding tasks draw more heavily on limited capacity [103, 104]. The fatigue factor reduces throughput multiplicatively [10, 12]. How the work rate drives task completion and deadline outcomes is formalised in Section 3.5.

Modelling choices. The model is deliberately parsimonious, but it is neither deterministic nor sinusoidal. Attention is perturbed by Gaussian process noise (σ_p in Eq. (3)). The circadian term is the sum of two Gaussian components (a late-morning peak and a post-lunch trough, Eq. (2)), as in the three-process model of alertness [14], rather than a sinusoid. Attention and fatigue are coupled through ϕ and jointly determine productivity through Eq. (5). Each functional form has a direct counterpart in the literature: exponential depletion and recovery in fatigue–recovery models [30, 31], resource depletion in vigilance research [7, 8] and additive homeostatic and circadian components in biomathematical alertness models [14, 112]. Individual variability is represented in three ways. Initial states are random. The rate parameters are scaled for resilient and fatigue-prone profiles (Section 6.8). Every cognitive parameter is perturbed by ±30% in a sensitivity analysis (Section 6.9). The model is not calibrated to individual data, which we state as a limitation (Section 7.5).

Initial state and behavioural plausibility. Each day starts with $H_0 \sim \mathcal{U}[0.8, 1.0]$ and $F_0 \sim \mathcal{U}[0, 0.15]$. Parameters were chosen *a priori* to reproduce qualitative patterns reported in the time-on-task and fatigue literature rather than fitted to a particular dataset. Under continuous work without breaks, attention falls from ≈ 1.0 to ≈ 0.3 by the end of the day, with a visible early-afternoon trough, and fatigue rises to ≈ 0.85. One 10-minute break recovers 30% of the attentional deficit and 20% of accumulated fatigue (Fig. 1b,c). Because no single parameterisation can represent all workers, Section 6.8 evaluates the policies zero-shot on resilient and fatigue-prone profiles in which all four rate parameters are scaled by ±30%.

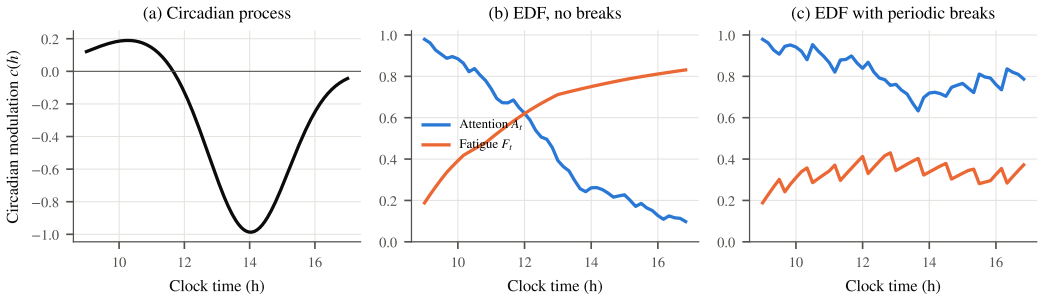


Figure 1: Simulator dynamics. (a) Circadian modulation $c(h)$ with a late-morning peak and post-lunch dip. (b) Effective attention and fatigue on one held-out test day under EDF without breaks. (c) The same day under EDF with a break after every four work slots (shaded bands mark breaks). Breaks keep attention high and fatigue low at the cost of working time.

3.3. Observation model and partial observability

The scheduler cannot observe H_t, A_t or F_t. It receives noisy estimates

$$\tilde{A}_t = \mathrm{clip}(A_t + \sigma_o \epsilon_t^A, 0, 1), \qquad \tilde{F}_t = \mathrm{clip}(F_t + \sigma_o \epsilon_t^F, 0, 1), \qquad \epsilon_t^A, \epsilon_t^F \sim \mathcal{N}(0,1), \tag{6}$$

with $\sigma_o = 0.10$ in the nominal setting. These estimates stand in for wearable, ocular, behavioural or self-report estimators, whose errors are substantial in practice [52–54]. The full observation is the 13-dimensional vector

$$o_t = \Big[\tfrac{t}{T},\ \tilde{A}_t,\ \tilde{F}_t,\ \underbrace{\tfrac{n_t^e}{N},\ \tfrac{W_t^e}{20},\ \sigma_t^e,\ \tfrac{m_t^e}{N},\ \tfrac{\omega_t^e}{4}}_{\text{easy class}},\ \underbrace{\tfrac{n_t^h}{N},\ \tfrac{W_t^h}{20},\ \sigma_t^h,\ \tfrac{m_t^h}{N},\ \tfrac{\omega_t^h}{4}}_{\text{hard class}} \Big], \tag{7}$$

where, for each class, n_t is the number of available unfinished tasks, W_t their remaining work, $\sigma_t = \mathrm{clip}\big((\min_i D_i - t)/T, -1, 1\big)$ the normalised slack of the most urgent task (1 if the class is empty), m_t the number of overdue tasks, and ω_t the remaining work of the shortest task (0 if the class is empty). Unlike a binary completion vector, this summary is invariant to the number of tasks and exposes the deadline information that urgency-aware scheduling requires. The worker's

true state and the future arrivals are hidden, so the decision problem is a POMDP $\langle \mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma_{\mathrm{RL}} \rangle$ [59]. Here $\mathcal{S}$ contains the full task set and (H_t, F_t, t), Ω is the observation space of Eq. (7), O is given by Eq. (6), and P by Eqs. (2)–(5) together with the arrival process.

3.4. Action space

The agent chooses among five macro-actions,

$$\mathcal{A} = \{\textsc{easy},\ \textsc{hard},\ \textsc{edf},\ \textsc{spt},\ \textsc{break}\}. \tag{8}$$

EASY (HARD) dispatches the available easy (hard) task with the earliest deadline, falling back to the other class if the chosen class is empty. EDF and SPT dispatch, across both classes, the task with the earliest deadline or the least remaining work. BREAK allocates the slot to rest, and the worker also rests when no task is available. The first two actions let the agent regulate the cognitive *intensity* of work relative to the worker's state. The two rule actions let it switch between urgency-driven and throughput-driven sequencing, in the spirit of DRL schedulers that select among dispatching rules [43, 48]. This hybrid design has an important methodological consequence: every rule-based baseline in Section 5.1 can be expressed as a policy over $\mathcal{A}$, so a learned policy is never handicapped relative to the baselines by its action space. Section 6.6 reports an agent restricted to the intensity actions $\{\textsc{easy}, \textsc{hard}, \textsc{break}\}$ of the original formulation. The action space is independent of N, and tasks are preemptible at slot boundaries, which is common in knowledge work.

3.5. Decision epochs, task progression and deadlines

Decisions are taken at the start of every slot $t = 0, \ldots, T-1$. Let $\mathcal{P}_t = \{i : e_i \leq t,\ \tau_i > t\}$ be the set of available unfinished tasks, where τ_i is the completion slot of

task i (initially ∞), and let $R_{i,t}$ be its remaining work ($R_{i,0} = w_i$). A macro-action $a_t \in \mathcal{A}$ is mapped to a task $j_t = g(a_t, \mathcal{P}_t) \in \mathcal{P}_t \cup \{\varnothing\}$ by the dispatch function

$$g(a, \mathcal{P}) = \begin{cases} \arg\min_{i \in \mathcal{P} \cap \mathcal{E}} D_i & a = \textsc{easy} \ \ (\text{or } \arg\min_{i \in \mathcal{P}} D_i \text{ if } \mathcal{P} \cap \mathcal{E} = \varnothing), \\ \arg\min_{i \in \mathcal{P} \cap \mathcal{H}} D_i & a = \textsc{hard} \ \ (\text{or } \arg\min_{i \in \mathcal{P}} D_i \text{ if } \mathcal{P} \cap \mathcal{H} = \varnothing), \\ \arg\min_{i \in \mathcal{P}} D_i & a = \textsc{edf}, \\ \arg\min_{i \in \mathcal{P}} R_{i,t} & a = \textsc{spt}, \\ \varnothing & a = \textsc{break} \text{ or } \mathcal{P} = \varnothing, \end{cases} \tag{9}$$

where $\mathcal{E}$ and $\mathcal{H}$ are the easy and hard tasks and ties are broken by task index. If $j_t \neq \varnothing$ the worker works ($u_t = 1$, $b_t = 0$) and the remaining work of the selected task decreases by the current work rate:

$$R_{j_t,t+1} = \max\{0,\ R_{j_t,t} - \rho_t\}, \qquad \tau_{j_t} = t + 1 \ \text{ if } R_{j_t,t+1} = 0, \tag{10}$$

while all other tasks are unchanged. Otherwise the worker rests ($u_t = 0$, $b_t = 1$). The cognitive state is then updated with Eqs. (3)–(4), the clock advances to $t + 1$, and the sets of tasks completed and deadlines missed in slot t are

$$C_t = \{i : \tau_i = t + 1\}, \qquad \mathcal{M}_t = \{i : D_i = t + 1,\ \tau_i > t + 1\}. \tag{11}$$

A task completed after its deadline counts as late, and a task still unfinished at $t = T$ has necessarily missed its deadline, because $D_i \leq T$. Preemption is allowed at slot boundaries, so the agent can interrupt a long task to insert a break or an urgent task. These rules make the interaction between scheduling decisions, time progression and deadline violations fully explicit. The pseudo-code of one slot is given in Supplementary Section S1.

3.6. Reward

The per-slot reward combines task outcomes with cognitive terms:

$$r_t = \underbrace{\sum_{i \in C_t} (1 + d_i) \left[\mathbb{1}(t + 1 \le D_i) + 0.25\, \mathbb{1}(t + 1 > D_i) \right] - 0.5\, |\mathcal{M}_t|}_{r_t^{\text{task}}} + \mu A_{t+1} - \lambda F_{t+1}, \tag{12}$$

where C_t is the set of tasks completed in slot t, $\mathcal{M}_t$ is the set of tasks whose deadline expires unfinished at the end of slot t, $\mu = 0.05$ and $\lambda = 0.10$. Harder tasks are worth more, late completion retains a quarter of the value, and each missed deadline costs 0.5. The attention and fatigue terms express the organisation's preference for sustainable work. Because such terms can change the optimal policy [146], we (i) report reward-independent operational metrics (on-time rate, completion, fatigue exposure) alongside the return, (ii) train an agent with $\lambda = \mu = 0$ to show that the learned behaviour is driven by the dynamics rather than by the reward terms, and (iii) sweep λ to trace the productivity–fatigue Pareto front. The objective is to find a policy π that maps observation histories to actions and maximises the expected discounted return

$$J(\pi) = \mathbb{E}_\pi \Big[\sum_{t=0}^{T-1} \gamma_{\text{RL}}^t\, r_t \Big]. \tag{13}$$

Table 2: Nominal simulator parameters and their rationale.

Parameter	Value	Rationale / source
Slots per day T, slot length Δ	48, 10 min	8-h workday; decision granularity of micro-breaks [22]
Tasks per day N (initial / arriving)	12 (6 / 6)	conditionally overloaded day; stochastic arrivals
Difficulty d_i, hard threshold	$\mathcal{U}[0.2, 1]$, 0.6	range of the original formulation
Work content w_i	$1 + 2.5d_i$ units	15–35 min at full capacity
Relative slack s_i	$\mathcal{U}[0.3, 1]$	range of the original formulation
Attention depletion α, fatigue coupling ϕ	0.02, 0.5	vigilance decrement over hours of work [7, 8]
Attention recovery β	0.30 per slot	break benefits [22, 106]
Circadian amplitude γ	0.15	post-lunch dip [15, 16]
Fatigue accumulation δ, recovery η	0.05, 0.20	exponential fatigue–recovery [30, 31]
Fatigue throughput penalty κ	0.5	performance loss under fatigue [10, 150]
Process / observation noise σ_p, σ_o	0.02, 0.10	variability; sensor error [52, 54]
Reward weights μ, λ	0.05, 0.10	swept in Section 6.5

4. Attention-aware deep reinforcement learning

The framework has three learner-independent elements: the observation history that approximates the belief state, the action sets of Section 3.4 and the reward of Eq. (12). We instantiate it with a value-based learner (AA-DQN, Sections 4.1–4.3) and a policy-gradient learner (AA-PPO, Section 4.4). Figure 2 gives an overview of the closed loop between the simulated workday and the learning agent.

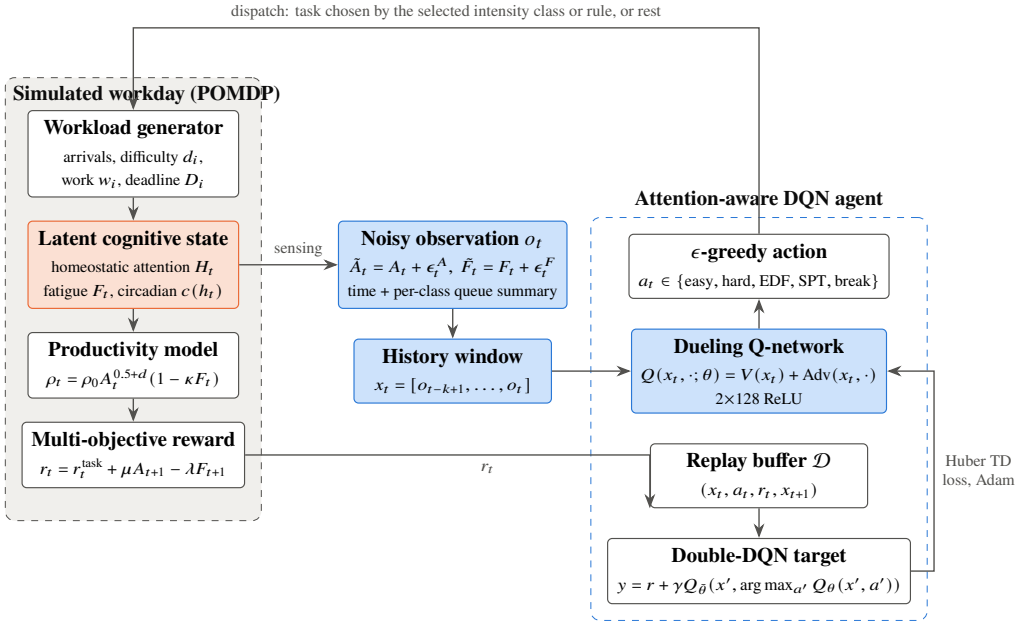


Figure 2: Overview of the attention-aware scheduling framework. The worker's cognitive state is latent; the agent receives noisy estimates together with a queue summary, encodes a short observation history, and selects an intensity class, a dispatching rule or a break. In AA-DQN (shown), transitions are stored in a replay buffer and used to train a dueling Q-network with Double-DQN targets; in AA-PPO, the same history feeds an actor–critic network trained on on-policy rollouts.

4.1. From Q-learning to deep Q-networks

For a fully observable MDP, the optimal action-value function satisfies the Bellman optimality equation [151, 152]

$$Q^*(s,a) = \mathbb{E}\big[r + \gamma_{\mathrm{RL}} \max_{a'} Q^*(s',a') \,\big|\, s,a\big], \tag{14}$$

and tabular Q-learning converges to Q^* under standard conditions [96]. With continuous cognitive variables, a tabular representation is infeasible, so Q is approximated by a neural network $Q(\cdot,\cdot;\theta)$ trained to minimise the temporal-difference (TD) error. The DQN of Mnih et al. [97] stabilises this nonlinear approximation with an experience replay buffer [141], which breaks the temporal correlation of samples, and a periodically synchronised target network $\bar{\theta}$. Convergence is not guaranteed for nonlinear function approximation [143], so we assess stability empirically across independent seeds.

4.2. Handling partial observability with an observation history

In a POMDP, the optimal policy is a function of the belief [57, 59]

$$\mathcal{B}_t(s) = \Pr\big(s_t = s \mid o_{0:t}, a_{0:t-1}\big). \tag{15}$$

Exact belief tracking is intractable for our continuous, nonlinear dynamics, and a memoryless policy $\pi(o_t)$ can be arbitrarily suboptimal [64]. Two structural properties of the scheduling problem make a short history informative. First, the latent state is persistent. H_t and F_t change slowly relative to the observation noise, so averaging several consecutive estimates reduces the error: for k independent observations of a quasi-static quantity, the standard error falls by a factor $\sqrt{k}$. Second, the trajectory of the observations, for example fatigue rising after consecutive hard tasks or attention recovering after a break, reveals the direction and rate of

change, which a single noisy reading cannot. We therefore feed the network the window

$$x_t = [o_{t-k+1}, \dots, o_t] \in \mathbb{R}^{13k}, \qquad k = 4, \tag{16}$$

zero-padded at the start of the day. This is the finite-memory approximation used for Atari [97]. It is computationally cheaper than recurrent alternatives [65, 140] and is a strong baseline for many POMDPs [66]. A 40-minute window spans the time scale of a typical work bout. Section 6.6 varies $k \in \{1, 2, 4, 8\}$.

4.3. Double Q-learning and dueling architecture

Max-based targets overestimate action values in noisy environments [67], and our environment is noisy by design. AA-DQN therefore uses the Double-DQN target, which decouples action selection (online network) from evaluation (target network):

$$y_t = r_t + \gamma_{\mathrm{RL}}(1 - done_t)\, Q\big(x_{t+1},\ \arg\max_{a'} Q(x_{t+1}, a'; \theta);\ \bar{\theta}\big). \tag{17}$$

In many states, for example at the start of the day or when the queue is empty, the choice between easy and hard work matters much less than the value of the state itself. A dueling head [68] represents this explicitly:

$$Q(x, a; \theta) = V(x; \theta) + \Big(\mathrm{Adv}(x, a; \theta) - \tfrac{1}{|\mathcal{A}|} \textstyle\sum_{a'} \mathrm{Adv}(x, a'; \theta)\Big). \tag{18}$$

The shared body consists of two fully connected layers of 128 ReLU units. The loss is the Huber loss of the TD error, which is robust to the occasional large reward of a hard task completed on time [153]:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,a,r,x')\sim\mathcal{D}}\big[\ell_{\delta=1}\big(Q(x, a; \theta) - y\big)\big], \quad \ell_\delta(u) = \begin{cases} \tfrac{1}{2}u^2, & |u| \le \delta, \\ \delta\big(|u| - \tfrac{1}{2}\delta\big), & \text{otherwise.} \end{cases} \tag{19}$$

Parameters are updated with Adam [154] and gradient-norm clipping. The vanilla DQN baseline uses the same action set and observation but $k = 1$, the standard max target and a single linear output head. This isolates the contribution of memory and of the two stabilisation components.

4.4. Policy-gradient instantiation (AA-PPO)

The framework does not depend on a particular learning algorithm: the observation history, the action sets and the reward are shared, and only the learner changes. To test whether a policy-gradient learner exploits the same information better than a value-based one, we instantiate the framework with proximal policy optimisation [69] and call the result AA-PPO. A stochastic policy $\pi_\theta(a \mid x)$ and a value function $V_\psi(x)$ share two hidden layers and are trained on rollouts of on-policy experience. Each update maximises the clipped surrogate objective

$$\begin{aligned} \mathcal{L}^{\mathrm{PPO}}(\theta) = \mathbb{E}_t\Big[\min\big(\varrho_t(\theta)\widehat{\mathrm{Adv}}_t,\ \mathrm{clip}(\varrho_t(\theta), 1-\varepsilon_{\mathrm{c}}, 1+\varepsilon_{\mathrm{c}})\widehat{\mathrm{Adv}}_t\big)\Big] \\ - c_v\, \mathbb{E}_t\big[(V_\psi(x_t) - \hat{G}_t)^2\big] + c_e\, \mathbb{E}_t\big[\mathbb{H}(\pi_\theta(\cdot \mid x_t))\big], \end{aligned} \tag{20}$$

where $\varrho_t(\theta) = \pi_\theta(a_t \mid x_t)/\pi_{\theta_{\mathrm{old}}}(a_t \mid x_t)$ is the probability ratio, $\widehat{\mathrm{Adv}}_t$ the generalised advantage estimate [155], $\hat{G}_t = \widehat{\mathrm{Adv}}_t + V_\psi(x_t)$ the return target, $\mathbb{H}$ the entropy, and $\varepsilon_{\mathrm{c}} = 0.2$, $c_v = 0.5$, $c_e = 0.01$. At test time AA-PPO acts greedily, $a_t = \arg\max_a \pi_\theta(a \mid x_t)$. AA-PPO was added during the revision of this paper in response to a reviewer's request for an alternative DRL method. Its settings are the standard defaults and were not tuned (Section 5.1).

4.5. Exploration and training procedure

Actions are selected ϵ-greedily with an exponentially decaying exploration rate

$$\epsilon_n = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})\, e^{-\kappa_\epsilon n}, \tag{21}$$

where n is the environment step, $\epsilon_{\max} = 1$ and $\epsilon_{\min} = 0.01$, and κ_ϵ is set so that ϵ reaches 0.05 after half of the training budget. Algorithm 1 summarises training. Every 100 episodes, the greedy policy is evaluated on a fixed set of 100 *validation* days that are disjoint from the training and test days, and the parameters with the highest validation return are retained. This mirrors the tuning of the heuristic baselines on their own held-out days (Section 5.1), so that both families receive the same model-selection budget and neither sees the test days. For transparency we also report the final-iterate policy (Supplementary Table S2).

4.6. Computational complexity

A forward pass costs $O(13k \cdot 128 + 128^2 + 128 \cdot |\mathcal{A}|) \approx 2.4 \times 10^4$ multiply–accumulate operations, independent of the number of tasks, because the observation summarises the queue per class. The dispatch step is $O(N \log N)$. Inference therefore takes microseconds on a CPU (Section 6.10), which is negligible relative to a 10-minute decision interval and allows deployment on wearable or edge devices.

Algorithm 1 Training of the attention-aware DQN (AA-DQN).

Require: episodes E, history length k, replay capacity $|\mathcal{D}|$, batch size B, target period C, validation days $\mathcal{V}$

initialise θ randomly, $\bar{\theta} \leftarrow \theta$, $\mathcal{D} \leftarrow \emptyset$, $n \leftarrow 0$, $J^\star \leftarrow -\infty$
for episode $e = 1, \dots, E$ **do**
    sample a workday (tasks, initial state, noise); observe o_0; $x_0 \leftarrow [\mathbf{0}, \dots, \mathbf{0}, o_0]$
    for $t = 0, \dots, T-1$ **do**
        $a_t \leftarrow$ random action with prob. ϵ_n, else $\arg\max_a Q(x_t, a; \theta)$
        dispatch $j_t = g(a_t, \mathcal{P}_t)$ (Eq. (9)); simulate Eqs. (2)–(5); receive r_t (Eq. (12)) and o_{t+1}
        $x_{t+1} \leftarrow \mathrm{shift}(x_t, o_{t+1})$; store $(x_t, a_t, r_t, x_{t+1}, done_t)$ in $\mathcal{D}$; $n \leftarrow n+1$
        if $|\mathcal{D}| \geq$ warm-up **then**
            sample a minibatch of B transitions; compute targets with Eq. (17)
            $\theta \leftarrow \theta - \eta_{\mathrm{lr}} \,\mathrm{Adam}\big(\nabla_\theta \mathcal{L}(\theta)\big)$ with gradient clipping
        if $n \bmod C = 0$ **then** $\bar{\theta} \leftarrow \theta$
    if $e \bmod 100 = 0$ **then**
        $J \leftarrow$ mean greedy return on $\mathcal{V}$; **if** $J > J^\star$ **then** $J^\star \leftarrow J$, $\theta^\star \leftarrow \theta$
return $\theta^\star$

5. Experimental design

5.1. Baselines

We compare the two learners of the framework, AA-DQN and AA-PPO, against nine rule-based schedulers, which span classical dispatching and hand-crafted cognition-aware policies, and against a vanilla, memoryless DQN. Baselines dispatch individual tasks directly, so they have at least as much control over task choice as the RL agents.

- **Random**: uniform choice among EASY, HARD and BREAK.
- **FIFO**: the available task that arrived first; never rests.
- **EDF**: the available task with the earliest deadline; never rests [2, 5].
- **SPT**: the available task with the least remaining work; never rests [1].
- **EDF-Periodic**: EDF with one break after every p work slots, a fixed work–rest cycle in the spirit of ergonomic rest-allowance guidance [34].
- **EDF-Threshold**: EDF that rests whenever the *observed* fatigue exceeds τ_F or the observed attention falls below τ_A. This is a reactive, state-aware rule.
- **SPT-Periodic** and **SPT-Threshold**: the same two rest rules combined with SPT sequencing. SPT limits the damage of overload by finishing short tasks first, so these variants combine the best classical sequencing rule (Section 6.2) with state-aware rest. They were added after a pre-submission review identified them as the strongest hand-designed competitors (Supplementary Section S7).
- **Cog-Heuristic**: rests on the same threshold rule and otherwise schedules a hard task only when observed attention is at least τ_H, an easy task otherwise (EDF within class). This hand-crafted rule encodes the intuition that the learned policy might discover.

Policy-gradient learner. AA-PPO (Section 4.4) uses the same observation history ($k = 4$), the same action sets (hybrid and intensity-only, ten seeds each), the same training budget and the same validation-based checkpoint selection as AA-DQN. Its actor and critic share two hidden layers of 128 tanh units. The other

settings are the common defaults: rollouts of 2048 steps, 10 epochs, minibatches of 256, clipping 0.2, GAE parameter 0.95, entropy coefficient 0.01 and learning rate 3×10^{-4}.

The parameters of the five parametric heuristics were selected by exhaustive grid search, maximising mean return on 300 *tuning* days that are disjoint from the training, validation and test days. The grids were $p \in \{2, 3, 4, 5, 6, 8, 10\}$; $\tau_F \in \{0.3, \ldots, 0.9, 1.01\}$ and $\tau_A \in \{0, 0.2, \ldots, 0.7\}$ (56 combinations, for both threshold rules); and, for Cog-Heuristic, $\tau_F \times \tau_A \times \tau_H$ with $\tau_H \in \{0.4, \ldots, 0.9\}$ (252 combinations). The heuristics observe exactly the same noisy signals as the agents. The selected parameters are reported in Supplementary Table S1. Tuning is repeated separately for every environment variant, so that each ablation compares the agent with the best heuristic *for that variant*.

5.2. Protocol

The main configurations were trained with ten independent seeds, the environment ablations and the domain-randomised variant with five, and the reward-weight sweep and the component study with three, each for 5000 episodes (2.4×10^5 decisions). Four disjoint sets of days were used: training days (generated from the training seed), 100 validation days for checkpoint selection, 300 tuning days for heuristic parameters, and 500 test days used only once, for the final evaluation. All methods are evaluated greedily on the *same* 500 test days with *common random numbers*. Each test day fixes the task set, the initial cognitive state and the sequences of process and observation noise, so differences between methods are paired and not confounded by sampling variation. Seven further seeds of the intensity-only variant were trained for the secondary analysis (Section 6.7). During the revision, ten seeds of each AA-PPO variant and three seeds of each of

four hyperparameter settings of AA-DQN were added. In total, 114 agents were trained and more than 10^6 test days were simulated.

5.3. Metrics

Besides the return (Eq. (13) with $\gamma_{\mathrm{RL}} = 1$), which depends on the reward weights, we report operational metrics that are independent of the reward and are computed from the task outcomes and the *true* latent states. With τ_i the completion slot of task i (Eq. (10)), the on-time rate, the difficulty-weighted on-time rate and the completion rate are

$$\mathrm{OT} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(\tau_i \le D_i), \quad \mathrm{WOT} = \frac{\sum_i d_i \, \mathbb{1}(\tau_i \le D_i)}{\sum_i d_i}, \quad \mathrm{CR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(\tau_i \le T), \tag{22}$$

so that $1 - \mathrm{OT}$ is the deadline-violation rate and WOT measures workload-adjusted productivity. The mean attention, mean fatigue, high-fatigue exposure and break share are

$$\bar{A} = \frac{1}{T}\sum_{t=1}^{T} A_t, \quad \bar{F} = \frac{1}{T}\sum_{t=1}^{T} F_t, \quad \mathrm{HFE} = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}(F_t > 0.7), \quad \mathrm{BS} = \frac{1}{T}\sum_{t=0}^{T-1} b_t. \tag{23}$$

Because the reward contains A_t and F_t (Eq. (12)), claims about cognitive sustainability are based on the reward-independent outcomes OT, WOT and CR, and are checked with an agent trained without cognitive reward terms (Section 6.4).

5.4. Statistical analysis

Following recommendations for reliable RL evaluation [60, 61], we treat the training seed as the unit of replication for learned policies. For each seed, metrics are averaged over the 500 test days. We report the mean and standard deviation across seeds, and 95% bootstrap confidence intervals [156] with 5000 resamples.

Deterministic heuristics are summarised across test days. To compare a learned agent with a baseline, we average the agent over its seeds for each test day and apply the two-sided Wilcoxon signed-rank test [157] to the 500 paired day-level differences. We report the matched-pairs rank-biserial correlation r_{rb} as effect size, the win rate, and Holm-adjusted p-values [158] over the nine baseline comparisons. As a more conservative seed-level check, we also report how many of the ten seeds outperform each baseline.

5.5. How the parameters were set

Parameters fall into three groups, and each was set without using the test days. (i) *Simulator parameters* (Table 2) were fixed a priori to reproduce qualitative patterns from the time-on-task and fatigue literature (Section 3). Their influence is quantified by a one-at-a-time ±30% sensitivity analysis and by worker-profile tests (Sections 6.9 and 6.8). (ii) *Agent hyperparameters* (Table 3) are standard values from the DQN and PPO literature [67–69, 97] and were not tuned on test data. Their sensitivity is reported for the learning rate, the discount factor, the history length and each architectural component (Section 6.9). (iii) *Heuristic parameters* were tuned by grid search on dedicated tuning days, as described above. The training budget (5000 episodes) and the action set were fixed once, before the final experiments, as documented in Supplementary Section S7.

5.6. Implementation

The simulator, agents and baselines are implemented in Python with NumPy and PyTorch [159]. Table 3 lists the hyperparameters. Training used one CPU thread per run. The code, configuration files, raw per-day results and analysis

scripts are released (see Data availability) so that every number and figure in this paper can be regenerated with a single command.

Table 3: Agent hyperparameters. AA-PPO uses the same observation history, action sets, training budget and checkpoint selection as AA-DQN.

Hyperparameter	DQN	AA-DQN
History length k	1	4
Target	max (standard)	Double DQN
Output head	linear	dueling (V + advantage)
Hidden layers	2×128, ReLU	
Optimiser, learning rate	Adam, 5×10^{-4}	
Discount γ_{RL}	0.99	
Replay capacity, warm-up	50 000, 1 000 transitions	
Minibatch size	64	
Target update period C	1 000 steps (hard copy)	
Loss, gradient clip	Huber ($\delta = 1$), $\|\nabla\|_2 \le 10$	
Exploration	ϵ: $1.0 \to 0.01$, exponential (0.05 at half budget)	
Action set	{EASY, HARD, EDF, SPT, BREAK}	
Training budget	5 000 episodes = 240 000 decisions	
Checkpoint selection	best greedy return on 100 validation days (every 100 ep.)	
AA-PPO		
Network	shared 2×128 tanh body; linear policy and value heads	
Optimiser, learning rate	Adam ($\epsilon = 10^{-5}$), 3×10^{-4}	
Rollout, epochs, minibatch	2 048 steps, 10, 256	
Clip ε_{c}, GAE λ_{GAE}, discount	0.2, 0.95, 0.99	
Value/entropy coef., grad. clip	0.5 / 0.01, $\|\nabla\|_2 \le 0.5$	
Action at test time	$\arg\max_a \pi_\theta(a \mid x)$ (greedy)	

6. Results

6.1. Learning dynamics

Figure 3 shows greedy validation performance during training, averaged over ten seeds. AA-DQN improves rapidly during the first 1000 episodes and slowly thereafter. The seed-averaged curve of its *current* parameters remains below the best tuned rule (SPT-Threshold) throughout training. This average hides substantial variation between checkpoints within each run: the best checkpoints of every seed reach the rule's level, while neighbouring checkpoints fall below it. Such policy oscillation is typical of value-based deep RL with a constant learning rate [60], and it motivates selecting checkpoints on held-out validation days. AA-PPO learns more slowly at first, overtakes AA-DQN after about 1700 episodes, exceeds the best rule on validation days after about 3000 episodes and is still improving at the end of the budget. Its learning curve is smooth: the final iterate is almost as good as the validation-selected checkpoint (return 13.02 versus 13.04), whereas for AA-DQN checkpoint selection matters (11.55 versus 12.17; Supplementary Table S2). Validation-selected checkpoints were taken at episode 3350 on average for AA-DQN, 4810 for AA-PPO and 3490 for DQN. Because validation and test days are disjoint, the test results below are not inflated by this selection. The vanilla DQN learns more slowly and plateaus well below AA-DQN, consistent with the over-estimation that Double Q-learning mitigates [67] and with the known weakness of memoryless policies in POMDPs [64].

6.2. Comparison with baselines

Table 4 and Figs. 4–5 report performance on the 500 held-out test days. Table 5 gives the paired statistical comparisons for AA-PPO with the hybrid action set; the

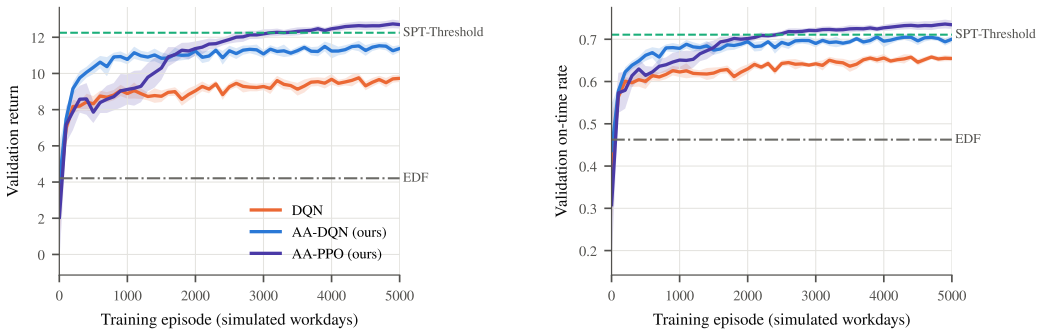


Figure 3: Greedy validation performance of the current parameters during training (mean of ten seeds; shaded bands are 95% confidence intervals). Horizontal lines show the best tuned rule (SPT-Threshold) and EDF on the same 100 validation days. Test results use the validation-selected checkpoint of each seed.

corresponding comparisons for AA-DQN are given in Supplementary Table S3. Four findings stand out.

(i) Capacity-agnostic rules fail under cognitive dynamics. The rules that never rest (FIFO, EDF, SPT) drive the worker into a low-capacity regime: mean fatigue reaches 0.48–0.55, attention falls to 0.46–0.54, and EDF completes only 46.8% of tasks on time. EDF performs particularly poorly, which reproduces the overload domino effect known from real-time scheduling [74]: once capacity falls, chasing the most urgent task means working on tasks that will be late anyway. SPT limits the damage by finishing short tasks first (62.5%).

(ii) State-aware rest is the dominant lever. Adding rest to these rules, either periodically or when observed fatigue or attention crosses a tuned threshold, raises the return by 4–8 units and roughly halves mean fatigue. All competitive policies rest in 15–20% of slots. The best rule overall is SPT-Threshold (return 12.30, on-time rate 71.4%), which combines SPT sequencing with threshold-triggered rest.

(iii) The learned policies match or exceed the best tuned rule, and AA-PPO exceeds it on every task-based metric. AA-PPO achieves a return of 13.04 ± 0.30 across seeds and an on-time rate of 75.1%. Relative to SPT-Threshold, the paired differences are +0.73 in return (95% CI [0.60, 0.87], $p_{\text{Holm}} < 10^{-4}$), +3.7 percentage points in on-time rate ([3.1, 4.3]) and +7.0 pp in difficulty-weighted on-time rate ([6.3, 7.7]), and all ten seeds are better than the rule on each of these metrics. AA-PPO also significantly exceeds every other rule in return and on-time rate ($p_{\text{Holm}} < 10^{-4}$), and it exceeds AA-DQN by 0.87 in return. With the intensity-only action set, AA-PPO is slightly better still (return 13.20, on-time rate 75.4%; +0.89 [0.73, 1.06] in return over SPT-Threshold) and has the smallest seed-to-seed variability of all learners (s.d. 0.10). AA-DQN with the hybrid action set ties with SPT-Threshold in return ($\Delta = -0.13$, [-0.29, 0.02], $p_{\text{Holm}} = 0.24$) but completes more tasks on time (72.8%; +1.4 pp, $p_{\text{Holm}} = 0.00016$).

The difficulty-weighted on-time rate reveals why the learned policies are better (Fig. 5). SPT-based rules complete many tasks on time by finishing short tasks first, but they neglect hard tasks (weighted on-time rate 59.7%). EDF-based rules protect hard tasks (65.7%) but complete fewer tasks overall. The learned policies combine the strengths of both. AA-PPO completes more tasks on time than any rule, while matching the EDF rules on difficulty-weighted completion (66.7%; the difference from EDF-Periodic, +1.1 pp, is not significant, $p_{\text{Holm}} = 0.062$). In other words, the learned policies avoid the starvation of long tasks that is the classical weakness of SPT [4]. No single hand-designed rule in our set achieves this.

(iv) The gains are not bought with lower completion, but the learned policies keep the worker somewhat more fatigued than the best rule. AA-PPO completes almost as many tasks in total as SPT-Threshold (80.9% versus 81.7%, counting

late completions), so its higher on-time rate does not come from abandoning tasks. It rests in 16.2% of slots versus 18.7% for SPT-Threshold, and its mean fatigue is higher by 0.025 (0.294 versus 0.268). This is still 46% below EDF, and the mean fatigue stays below 0.5 in every slot of the day (Fig. 6). AA-DQN rests least among the attention-aware learners (15.0%) and has the highest fatigue of the learned policies (0.333). Section 6.5 shows that the fatigue level can be lowered through the reward weight without losing on-time performance.

Relative to the memoryless vanilla DQN, adding memory, Double Q-learning and the dueling head increases the return by 1.88 ([1.81, 1.96]), with a win on 98.0% of test days. The vanilla DQN ranks below five of the nine rules, so a naive application of DQN to this POMDP would wrongly suggest that learning is not competitive. We report the validation-selected policies as primary results because the rules received an equivalent selection step (grid search) on their own held-out days; the final-iterate results lead to the same ranking among the learners (Supplementary Table S2).

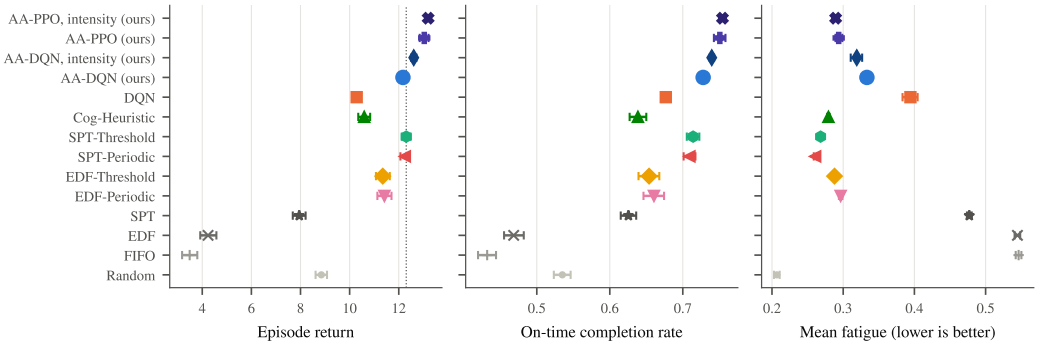


Figure 4: Test performance with 95% bootstrap confidence intervals (learned policies: across seeds; rules: across days). The dotted line marks the best rule's return.

Table 4: Performance on 500 held-out test days (nominal environment). Learned policies: mean ± s.d. across ten training seeds of the per-seed test means. Rules: mean ± s.d. across test days (they are deterministic given the day). Bold marks the best value in each column.

Method	Return	On-time rate	Weighted on-time	Mean attention	Mean fatigue	Break share
Random	8.85 ± 2.71	0.535 ± 0.131	0.488 ± 0.143	0.855 ± 0.037	0.207 ± 0.045	0.326 ± 0.065
FIFO	3.49 ± 3.49	0.432 ± 0.139	0.422 ± 0.125	0.463 ± 0.106	0.546 ± 0.045	0.000 ± 0.003
EDF	4.24 ± 3.82	0.468 ± 0.155	0.454 ± 0.137	0.465 ± 0.105	0.545 ± 0.046	0.000 ± 0.003
SPT	7.95 ± 2.94	0.625 ± 0.120	0.494 ± 0.122	0.543 ± 0.093	0.477 ± 0.038	0.000 ± 0.001
EDF-Periodic	11.41 ± 3.32	0.660 ± 0.160	0.657 ± 0.148	0.798 ± 0.040	0.296 ± 0.029	0.188 ± 0.005
EDF-Threshold	11.35 ± 3.28	0.654 ± 0.164	0.650 ± 0.155	0.806 ± 0.022	0.288 ± 0.034	0.201 ± 0.037
SPT-Periodic	12.25 ± 2.04	0.709 ± 0.096	0.591 ± 0.111	0.817 ± 0.034	0.261 ± 0.023	0.188 ± 0.005
SPT-Threshold	12.30 ± 2.03	0.714 ± 0.099	0.597 ± 0.114	0.813 ± 0.022	0.268 ± 0.036	0.187 ± 0.035
Cog-Heuristic	10.59 ± 2.76	0.638 ± 0.132	0.533 ± 0.150	0.803 ± 0.034	0.279 ± 0.020	0.198 ± 0.025
DQN	10.29 ± 0.26	0.677 ± 0.008	0.579 ± 0.012	0.677 ± 0.017	0.394 ± 0.018	0.108 ± 0.020
AA-DQN (ours, hybrid actions)	12.17 ± 0.15	0.728 ± 0.005	0.655 ± 0.012	0.754 ± 0.011	0.333 ± 0.012	0.150 ± 0.018
AA-DQN (ours, intensity actions)	12.61 ± 0.18	0.740 ± 0.005	0.678 ± 0.008	0.770 ± 0.012	0.319 ± 0.014	0.162 ± 0.016
AA-PPO (ours, hybrid actions)	13.04 ± 0.30	0.751 ± 0.013	0.667 ± 0.022	0.794 ± 0.007	0.294 ± 0.008	0.162 ± 0.012
AA-PPO (ours, intensity actions)	**13.20 ± 0.10**	**0.754 ± 0.003**	**0.689 ± 0.006**	0.800 ± 0.006	0.289 ± 0.008	0.179 ± 0.008

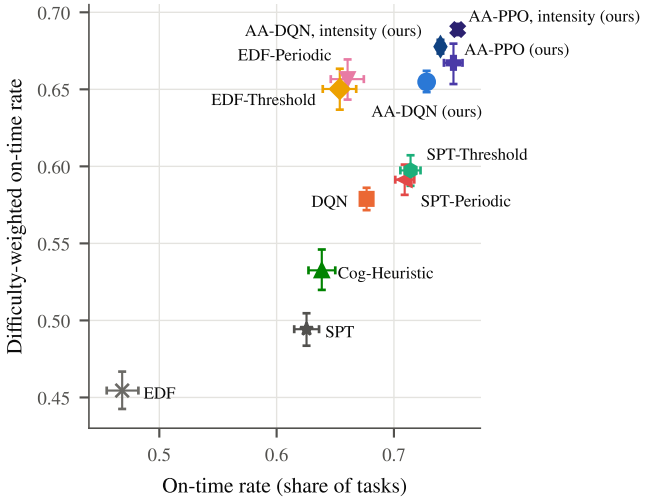


Figure 5: Count-based versus difficulty-weighted on-time completion (95% CIs). SPT-based rules complete many tasks but neglect hard ones; EDF-based rules protect hard tasks but complete fewer. The learned policies are the only ones near the best value on both axes.

Table 5: Paired comparison of AA-PPO (hybrid actions) with each baseline on 500 common-random-number test days: mean difference with 95% bootstrap CI (on-time rates in percentage points), matched-pairs rank-biserial correlation r_{rb} (positive favours AA-PPO), and Holm-adjusted Wilcoxon signed-rank p-values over the eleven comparisons.

	Episode return			On-time rate (pp)			Weighted on-time rate (pp)		
AA-PPO vs.	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}	Δ [95% CI]	r_{rb}	p_{Holm}
Random	+4.19 [+3.98, +4.39]	0.99	$< 10^{-4}$	+21.6 [+20.6, +22.6]	1.00	$< 10^{-4}$	+17.9 [+17.0, +18.9]	0.98	$< 10^{-4}$
FIFO	+9.55 [+9.32, +9.78]	1.00	$< 10^{-4}$	+31.9 [+31.0, +32.7]	1.00	$< 10^{-4}$	+24.5 [+23.8, +25.3]	1.00	$< 10^{-4}$
EDF	+8.80 [+8.56, +9.04]	1.00	$< 10^{-4}$	+28.2 [+27.3, +29.2]	1.00	$< 10^{-4}$	+21.3 [+20.5, +22.0]	1.00	$< 10^{-4}$
SPT	+5.09 [+4.92, +5.25]	1.00	$< 10^{-4}$	+12.5 [+11.9, +13.2]	0.99	$< 10^{-4}$	+17.3 [+16.6, +18.0]	1.00	$< 10^{-4}$
EDF-Periodic	+1.62 [+1.43, +1.81]	0.71	$< 10^{-4}$	+9.0 [+8.1, +9.9]	0.84	$< 10^{-4}$	+1.1 [+0.3, +1.8]	0.10	0.0624
EDF-Threshold	+1.69 [+1.50, +1.89]	0.73	$< 10^{-4}$	+9.7 [+8.7, +10.6]	0.86	$< 10^{-4}$	+1.7 [+0.9, +2.5]	0.15	0.0074
SPT-Periodic	+0.79 [+0.65, +0.93]	0.52	$< 10^{-4}$	+4.2 [+3.5, +4.8]	0.60	$< 10^{-4}$	+7.6 [+6.8, +8.4]	0.77	$< 10^{-4}$
SPT-Threshold	+0.73 [+0.60, +0.87]	0.51	$< 10^{-4}$	+3.7 [+3.1, +4.3]	0.58	$< 10^{-4}$	+7.0 [+6.3, +7.7]	0.77	$< 10^{-4}$
Cog-Heuristic	+2.44 [+2.29, +2.60]	0.95	$< 10^{-4}$	+11.2 [+10.5, +11.9]	0.98	$< 10^{-4}$	+13.5 [+12.7, +14.3]	0.96	$< 10^{-4}$
AA-DQN	+0.87 [+0.78, +0.95]	0.80	$< 10^{-4}$	+2.3 [+1.9, +2.7]	0.56	$< 10^{-4}$	+1.2 [+0.8, +1.7]	0.28	$< 10^{-4}$
AA-DQN, intensity	+0.43 [+0.32, +0.53]	0.38	$< 10^{-4}$	+1.1 [+0.7, +1.6]	0.22	$< 10^{-4}$	-1.1 [-1.5, -0.6]	-0.24	$< 10^{-4}$

6.3. What do the agents learn?

Figure 6 shows how actions and latent states evolve through the day, aggregated over the 500 test days (and over the ten seeds for the learned policies). Three behavioural regularities distinguish the learned policies from the rules.

Work while fresh. Both learners almost never rest in the first hour (AA-PPO 1.9% and AA-DQN 0.5% of slots, versus 3.6% for SPT-Threshold, 4.6% for EDF-Threshold and a fixed 16.7% for EDF-Periodic), and both devote about half of the first hour to hard tasks (AA-PPO 54%, AA-DQN 50%), when attention is highest. SPT-Threshold does almost no hard work in the first hour (8%), because hard tasks are long. A fixed cycle cannot distinguish a fresh worker at 09:30 from a recovered one at 15:00, and a threshold rule reacts only once a threshold has been crossed.

Rest when it is cheapest, and protect the middle of the day. AA-DQN's break share peaks at 24.6% between 13:00 and 14:00, which coincides with the

modelled post-lunch dip, and declines afterwards; in the 13:00–15:00 window it rests in 22.4% of slots, versus 12.5% at other times. Resting is cheapest when the productivity it forgoes is lowest (Eq. (5)). AA-PPO rests earlier and more evenly: 17% of slots between 10:00 and 11:00 and 24–27% between 11:00 and 14:00, before and during the dip, but only 9–11% after 15:00. As a result, its mean fatigue stays between 0.27 and 0.30 from 10:30 to 14:30, and mean attention never falls below 0.73. In the last two hours, AA-PPO works almost continuously to clear the remaining deadlines and lets fatigue rise to 0.45 at the end of the day. This is rational in a single-day model, in which end-of-day fatigue carries no future cost, but it would be penalised in a multi-day setting (Section 7.5). The threshold rules show a single, sharper peak (35–36% between 13:00 and 14:00) only because the dip pushes observed attention below their fixed threshold.

Match difficulty to capacity without starving hard tasks. The learned policies devote the most hard work to the morning and the least to the afternoon trough: AA-DQN goes from 50% at 09:00–11:00 to 33% at 13:00–15:00, and AA-PPO from 54% in the first hour to 18% between 13:00 and 14:00, filling the trough with easy tasks. Both return to hard work in the last hour (AA-DQN 48%, AA-PPO 78%) as the deadlines of the remaining hard tasks approach. SPT-Threshold postpones hard tasks until they become urgent and spends 82% of the last hour on hard work, when hard tasks are slowest (Eq. (5)), without having done hard work in the morning. This back-loading explains its low difficulty-weighted on-time rate (Section 6.2). EDF, in contrast, does more hard work in the afternoon (71%) than in the morning (60%) and never rests. The hand-crafted Cog-Heuristic was designed to capture the difficulty–attention intuition but, with a single attention threshold, also pushes hard tasks into the late afternoon (67% after 16:00). The learned policies combine

early hard work, easy work in the trough and late deadline-driven hard work with short-task completion, which illustrates how hard it is to encode such time-, state- and deadline-dependent trade-offs in rules.

Compared with the memoryless DQN, AA-DQN rests more (15.0% versus 10.8% of slots) and takes breaks at a lower true fatigue level (0.36 versus 0.44); AA-PPO rests at a still lower level (0.31). The DQN concentrates its rest in the last hour (29% of slots after 16:00), consistent with late, reactive breaks triggered by single noisy readings.

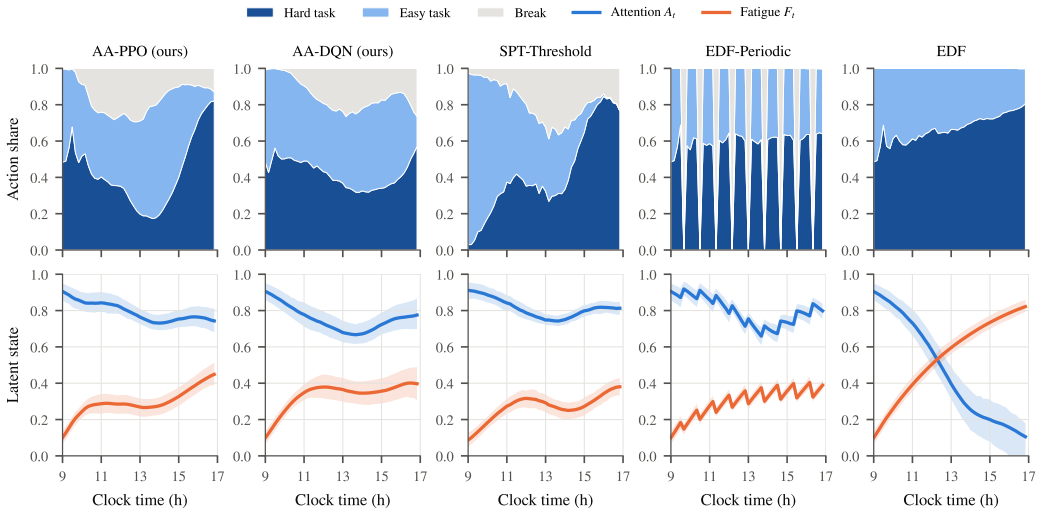


Figure 6: Learned and rule-based behaviour across the workday on 500 test days. Top: share of slots spent on hard tasks, easy tasks and breaks. Bottom: mean true attention and fatigue (bands: interquartile range). The saw-tooth pattern of EDF-Periodic reflects its fixed work–rest cycle.

6.4. Environment ablations: when does learning add value?

To separate the contributions of the environment's components, we retrained AA-DQN (five seeds) in four variants and re-tuned every heuristic for each variant. Table 6 and Fig. 7 report the difference between AA-DQN and the best heuristic

within each variant. This avoids comparing returns across variants with different reward scales. Each variant changes exactly one element of the environment, and both the agent and the rules are re-optimised for it, so the comparison is controlled. The ablations were run with AA-DQN, which was the primary learner when they were designed.

Observation quality is the largest single factor in this simulator. When attention and fatigue are observed without noise, AA-DQN improves to 13.61 and exceeds the best heuristic (SPT-Periodic, 12.25) by +1.36 ([1.19, 1.51]) in return and +5.5 percentage points in on-time rate. Its mean fatigue also falls, from 0.333 in the nominal setting to 0.269. The heuristics do not benefit from perfect information: the best threshold rule is no better than with noisy observations, because a fixed threshold cannot use precise state information beyond the threshold crossing. This suggests that the value of learning grows with sensor quality, whereas fixed-threshold rules cannot exploit more precise information.

The circadian process is not the source of the agent's advantage over rules. Without the circadian term, AA-DQN and the best rule are tied in return (+0.06, [-0.11, 0.23]), and AA-DQN keeps a +2.5 pp on-time advantage. This is the same pattern as in the full model, so the agent's behaviour around the post-lunch dip (Section 6.3) is an adaptation to the dynamics rather than the source of an advantage over the threshold rules.

The result does not depend on the cognitive reward terms. Trained with a task-only reward ($\lambda = \mu = 0$), AA-DQN still matches the best rule in return (-0.13, [-0.28, 0.03]) and still rests in 11.5% of slots (Table 7), because rest pays off through productivity alone. The agent's on-time advantage disappears in this variant (-0.3 pp), which indicates that the attention term shapes *how* it trades

completions against cognitive load.

Learning also helps without cognitive dynamics. In the static-capacity variant, in which rest has no value, AA-DQN beats the best rule (SPT) by +0.91 ([0.74, 1.07]) and +3.7 pp. Under overload, combining urgency (EDF) with processing time (SPT) depending on the state of the queue is better than either rule alone, a well-known property of composite dispatching [43, 76]. This result contradicts the claim, made in the original version of this manuscript, that reinforcement learning helps only when cognition is modelled, and we retract that claim. What cognitive dynamics change is the size of the stakes: in the full model, state-aware policies exceed the capacity-agnostic rules by 4–8 return units (Table 4), versus 1–3 units in the static model.

Summary. Within this simulator, modelling cognition is neither necessary for learning to help (static-capacity variant) nor sufficient for a clear advantage (nominal variant with noisy observations). The ablations identify what drives the results here: the quality of the state estimate, the rest decision and the integration of urgency and processing time. They do not establish necessity or sufficiency in general, which would require other simulators, action spaces and real data.

Table 6: Environment ablations. AA-DQN (five seeds) versus the best re-tuned heuristic within each variant; differences are paired over 500 test days with 95% bootstrap CIs (on-time in percentage points).

Environment variant	AA-DQN return	Best heuristic (return)	Δ return [95% CI]	AA-DQN on-time	Best heuristic (on-time)	Δ on-time [95% CI]
Full model	12.17 ± 0.15	SPT-Threshold (12.30)	-0.13 [-0.29, +0.02]	0.728	SPT-Threshold (0.714)	+0.014 [+0.007, +0.021]
No observation noise	13.61 ± 0.08	SPT-Periodic (12.25)	+1.36 [+1.19, +1.51]	0.764	SPT-Periodic (0.709)	+0.055 [+0.047, +0.062]
No circadian process	12.99 ± 0.21	SPT-Periodic (12.93)	+0.06 [-0.11, +0.23]	0.754	SPT-Threshold (0.729)	+0.025 [+0.017, +0.032]
Task-only reward	11.53 ± 0.18	SPT-Threshold (11.66)	-0.13 [-0.28, +0.03]	0.711	SPT-Threshold (0.715)	-0.003 [-0.011, +0.004]
Static capacity (no cognition)	10.41 ± 0.10	SPT (9.50)	+0.91 [+0.74, +1.07]	0.670	SPT (0.633)	+0.037 [+0.030, +0.045]

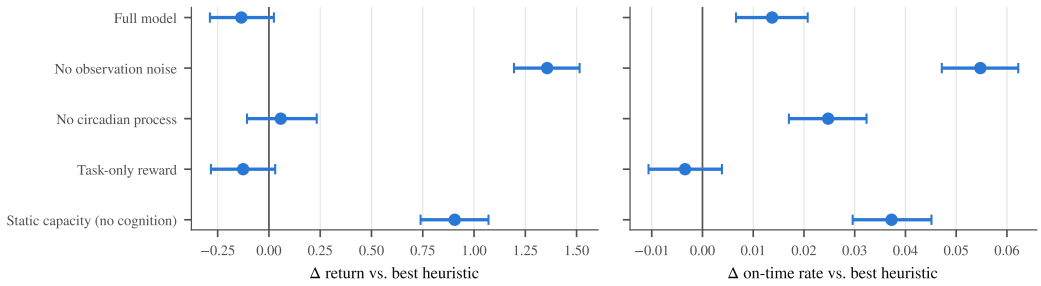


Figure 7: Difference between AA-DQN and the best heuristic within each environment variant (95% CIs). Positive values favour AA-DQN.

6.5. Productivity–fatigue trade-off

The fatigue weight λ (with $\mu = \lambda/2$) controls how much the agent values the worker's state. Figure 8 and Table 7 show the policies obtained for $\lambda \in \{0, 0.05, 0.1, 0.2, 0.4\}$. Increasing λ lowers mean fatigue monotonically, from 0.365 to 0.225, by raising the break share from 11.5% to 30.4%. Over most of this range it does *not* reduce productivity. The on-time rate rises from 70.8% ($\lambda = 0$) to 73.1% ($\lambda = 0.2$), and is still 72.3% at $\lambda = 0.4$. At $\lambda = 0.4$, AA-DQN keeps the worker *less* fatigued than any of the tuned heuristics (Fig. 8) while completing more tasks on time than the best of them (71.4%). Moderate concern for the worker's state is therefore not a cost to productivity but a means to it, which echoes empirical findings on rest breaks [18, 19, 22]. The fatigue penalty of the nominal agent noted in Section 6.2 is a consequence of the chosen weight, not of the method.

6.6. Component analysis

Table 8 isolates the components of AA-DQN (three seeds each, except for the ten-seed configurations); it also lists the hyperparameter variants discussed in Section 6.9 and the two AA-PPO configurations for reference. *Memory is*

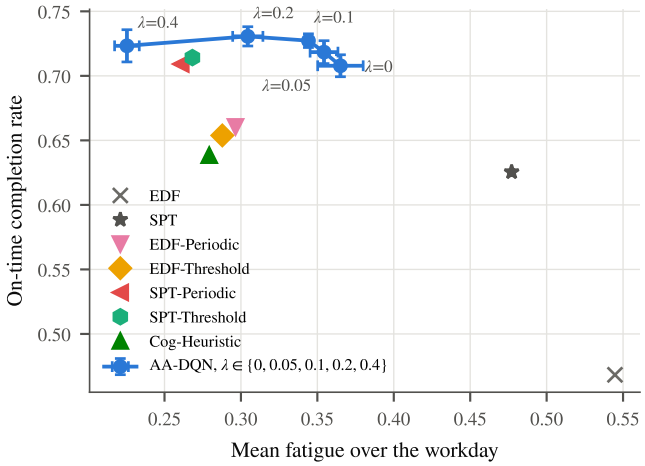


Figure 8: Productivity–fatigue trade-off. AA-DQN trained with different fatigue weights λ (three seeds each; error bars: s.d.) versus the tuned heuristics.

Table 7: Effect of the reward weight λ on AA-DQN (mean ± s.d. over three seeds; 500 test days).

λ ($\mu = \lambda/2$)	On-time rate	Mean fatigue	High-fatigue exposure	Break share
0	0.708 ± 0.009	0.365 ± 0.015	0.000	0.115
0.05	0.718 ± 0.009	0.354 ± 0.009	0.000	0.124
0.1	0.727 ± 0.005	0.344 ± 0.002	0.000	0.137
0.2	0.731 ± 0.007	0.304 ± 0.010	0.000	0.181
0.4	0.723 ± 0.012	0.225 ± 0.008	0.000	0.304

essential. Removing the observation history ($k = 1$) reduces the return from 12.17 to 10.34, close to the vanilla DQN, and a two-step history recovers only part of the loss (11.17). An eight-step history performs at least as well as $k = 4$ (12.48). A GRU encoder over an eight-step window performs similarly (12.54), so the benefit comes from the length of the memory rather than from recurrence per se. In contrast, Double Q-learning and the dueling head have small effects within seed variability (12.07 and 12.28 without them). This indicates that the partial observability of the cognitive state, not value over-estimation, is the main difficulty of the problem.

A smaller action space learns better. The agent restricted to the intensity actions {EASY, HARD, BREAK} achieved the highest return of all configurations in the three-seed study (12.61). Because this variant was not our pre-specified primary method, we did not promote it on the basis of three seeds. Instead, we re-evaluated it with the full ten-seed protocol as a secondary analysis (Section 6.7). The result is plausible in hindsight. With the shortest-task feature in the observation, choosing an easy (typically short) task already captures much of the benefit of SPT, while the larger hybrid action space slows exploration and credit assignment.

6.7. Secondary analysis: intensity-only action set

The component study (Section 6.6) suggested that restricting AA-DQN to the intensity actions {EASY, HARD, BREAK} improves performance. We therefore trained seven additional seeds of this variant and evaluated it with the same ten-seed protocol as the primary agent. Because the variant was selected after a look at test-set results from three seeds, we report it as a secondary, hypothesis-generating analysis. AA-PPO was trained with both action sets from the outset, so its intensity-only variant provides an independent check of this hypothesis.

Table 8: Component and sensitivity analysis (mean ± s.d. over seeds; 500 test days).

Configuration	Seeds	Return	On-time rate	Train time (s)
AA-DQN (hybrid actions, k=4, Double, Dueling)	10	12.17 ± 0.15	0.728 ± 0.005	720
– without history (k=1)	3	10.34 ± 0.19	0.683 ± 0.010	551
– without Double Q-learning	3	12.07 ± 0.19	0.720 ± 0.014	552
– without dueling head	3	12.28 ± 0.04	0.725 ± 0.004	517
– intensity-only actions {easy, hard, break}	10	12.61 ± 0.18	0.740 ± 0.005	582
history k=2	3	11.17 ± 0.12	0.701 ± 0.001	568
history k=8	3	12.48 ± 0.37	0.733 ± 0.011	605
GRU encoder over k=8 history	3	12.54 ± 0.16	0.728 ± 0.006	2463
learning rate 10^{-4} (nominal 5×10^{-4})	3	11.75 ± 0.16	0.719 ± 0.007	646
learning rate 10^{-3}	3	12.19 ± 0.17	0.726 ± 0.001	723
discount $\gamma_{\mathrm{RL}} = 0.95$ (nominal 0.99)	3	11.72 ± 0.29	0.712 ± 0.009	664
discount $\gamma_{\mathrm{RL}} = 0.999$	3	12.24 ± 0.08	0.727 ± 0.001	659
Vanilla DQN (k=1, no Double/Dueling)	10	10.29 ± 0.26	0.677 ± 0.008	614
AA-PPO (hybrid actions)	10	13.04 ± 0.30	0.751 ± 0.013	171
AA-PPO (intensity actions)	10	13.20 ± 0.10	0.754 ± 0.003	172

With ten seeds, the intensity-only AA-DQN achieves a return of 12.61 ± 0.18 and an on-time rate of 74.0%. Relative to the best rule, SPT-Threshold, the paired differences are 0.31 in return ([0.13, 0.48], $p_{\mathrm{Holm}} = 0.00016$), 2.5 pp in on-time rate ([1.8, 3.3], $p_{\mathrm{Holm}} < 10^{-4}$) and 8.0 pp in difficulty-weighted on-time rate ([7.1, 8.9]). Relative to the hybrid AA-DQN, the differences are 0.44 in return ([0.38, 0.50]) and 1.2 pp in on-time rate. The same ordering holds for AA-PPO, although the gap is smaller: the intensity-only variant reaches 13.20 versus 13.04 for the hybrid action set, with a lower seed-to-seed standard deviation (0.10 versus 0.30). Both learners therefore indicate that a compact, cognitively meaningful action space is at least as good as a larger rule-selection space for this problem: with

the shortest-task feature in the observation, choosing an easy (typically short) task already captures much of the benefit of SPT, while the larger action space slows exploration and credit assignment. Restricting the action space did not prevent the policies from respecting task-specific deadlines, because the task within each class is chosen by earliest deadline (Eq. (9)).

6.8. Robustness and generalisation

We evaluated all policies *zero-shot*, without retraining or re-tuning, under shifted conditions: different levels of sensor noise at test time, resilient and fatigue-prone worker profiles (all four rate parameters scaled by ±30%), and lighter or heavier workloads (Fig. 9; full results in Supplementary Table S5). We also trained five seeds of a domain-randomised agent (AA-DQN-DR) on randomly sampled workloads ($N \in \{8, \ldots, 16\}$), worker profiles (±30% on all rate parameters) and noise levels ($\sigma_o \in [0.05, 0.2]$).

Sensor noise. The learned policies are insensitive to the test-time noise level over the range they were trained near. The returns of both AA-DQN variants vary by less than 0.15 between $\sigma_o = 0$ and $\sigma_o = 0.3$, and those of AA-PPO by less than 0.05 up to $\sigma_o = 0.2$; at $\sigma_o = 0.3$, three times the training level, AA-PPO loses 0.3–0.5 but remains the best method. The threshold rules, in contrast, degrade as noise grows, because spurious threshold crossings trigger unnecessary breaks. SPT-Threshold falls from 12.30 to 11.35 and from 71.4% to 65.6% on-time, and EDF-Threshold falls from 11.35 to 9.63. The periodic rules are unaffected, because they ignore the cognitive signals altogether. The history-based policies average out noise that a threshold applied to a single reading cannot.

Worker profiles. AA-PPO is the best method for both profiles: for fatigue-prone workers its return is 11.04 (SPT-Threshold 10.59, AA-DQN 10.05), and for

resilient workers, who need less rest, the intensity-only AA-PPO reaches 15.46, ahead of the EDF-based rules (EDF-Periodic 15.01), which are the best rules for this profile. AA-DQN, in contrast, is overtaken by SPT-Threshold for fatigue-prone workers and by the EDF-based rules for resilient workers. The best rule depends on the profile, SPT-based for fatigue-prone and EDF-based for resilient workers, whereas the same AA-PPO policy is best for both.

Workload. Generalisation to a different number of tasks per day is the clearest weakness of AA-DQN and is much smaller for AA-PPO. With 16 tasks per day, both AA-PPO variants remain the best methods (11.83 versus 11.61 for SPT-Threshold), whereas AA-DQN falls well behind the SPT-based rules (9.69–10.03). With 8 tasks, the EDF-based rules complete almost all tasks on time and attain the highest return (EDF-Threshold 13.12); the intensity-only AA-PPO is close behind (13.04, with the same on-time rate of 97.4%), and AA-DQN falls behind (11.71–12.20). Although the observation is normalised by N, the queue statistics of these days lie outside the training distribution. Domain randomisation of AA-DQN improved the 16-task result (10.35) but did not close the gap and lowered nominal performance (11.63), so a simple randomisation over conditions with checkpoint selection on nominal days is insufficient. The better generalisation of AA-PPO suggests that the stochastic, on-policy learner finds smoother policies that depend less on the exact queue statistics, but we did not investigate the mechanism.

6.9. Sensitivity to hyperparameters and simulator parameters

This subsection quantifies how the results depend on the parameters that were fixed a priori (Section 5.5).

Agent hyperparameters. Table 8 reports AA-DQN for three learning rates, three discount factors, four history lengths, a recurrent encoder and with each

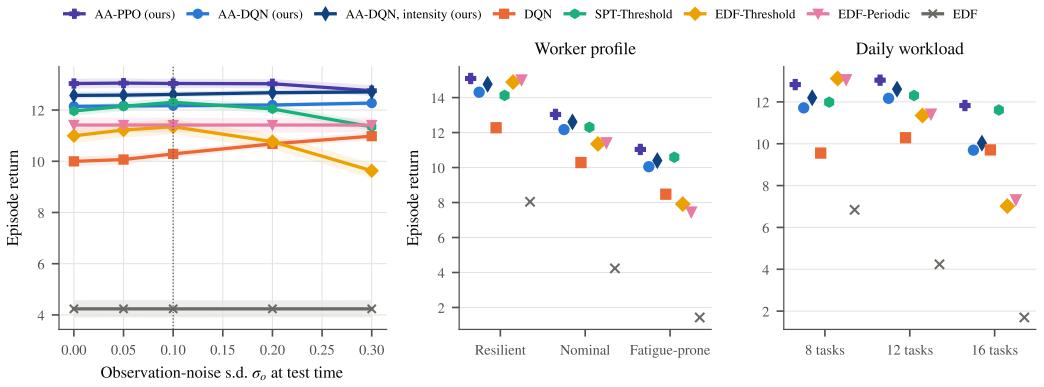


Figure 9: Zero-shot robustness. Left: return versus test-time observation noise (training/tuning at $\sigma_o = 0.1$, dotted line; shaded: 95% CIs). Centre and right: return for different worker profiles and daily workloads. Rules use parameters tuned in the nominal environment.

stabilisation component removed (three seeds each). Varying the learning rate from 5×10^{-4} to 10^{-3} or the discount factor from 0.99 to 0.999 changes the return by less than 0.1 (12.19 and 12.24 versus 12.17). Two settings are worse by about 0.45: a learning rate of 10^{-4} (11.75), which learns too slowly for the 5000-episode budget, and a discount factor of 0.95 (11.72), whose effective horizon of about 20 slots (3.3 h) is shorter than the day, so that consequences of decisions for late deadlines are discounted away. All these settings remain well above the vanilla DQN (10.29). The history length has the largest effect ($k = 1$: 10.34; $k = 8$: 12.48), as discussed in Section 6.6. AA-PPO used untuned default settings and nevertheless achieved the best results, so its performance is not the product of hyperparameter search. The reward weight λ is analysed in Section 6.5.

Simulator parameters. Each of the seven cognitive parameters (α, β, δ, η, κ, γ, ϕ) was decreased and increased by 30%, one at a time, and every trained policy and tuned rule was evaluated, without retraining or re-tuning, on the 500 test days under each of the 14 perturbed models (Table 9, Fig. 10). This is a conservative

test, because the learned policies and the rules are applied to dynamics they were not optimised for. The parameters that govern the depletion of attention (α), the accumulation of fatigue (δ) and its effect on throughput (κ) change the attainable return the most (by 1.2–1.4 for AA-PPO), whereas the circadian amplitude γ and the fatigue coupling ϕ have small effects. The fixed work–rest cycle of EDF-Periodic is the most sensitive policy (a range of 2.3 for α), because it cannot adapt the amount of rest. The ranking of the methods is remarkably stable. Both AA-PPO variants exceed the best rule in all 14 perturbed models (AA-PPO by at least 0.56; best learned policy minus best rule between 0.61 and 1.18), and the intensity-only AA-DQN does so in 12 of them. The hybrid AA-DQN, which ties with the best rule in the nominal model, falls slightly behind it in most perturbed models (2 of 14 wins; at worst by -0.52). The main conclusions therefore do not hinge on the particular parameter values in Table 2, although, as for any simulation study, they cannot replace calibration with real data (Section 7.5).

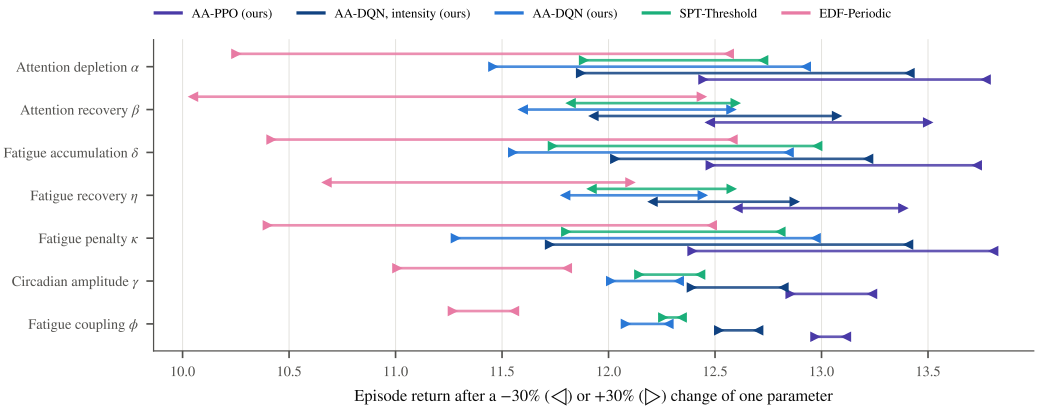


Figure 10: Return of selected policies after a −30% (◃) or +30% (▹) change of one cognitive parameter (500 test days, zero-shot). The learned policies stay ahead of the rules for every parameter.

Table 9: Zero-shot sensitivity to the cognitive parameters: return on 500 test days when one parameter is changed by ±30% (learned policies: mean of ten seeds; rules: parameters tuned in the nominal model). The last column gives the return of the best learned policy minus that of the best rule. Bold marks the best method in each row.

Parameter	Change	AA-PPO	AA-PPO, intensity	AA-DQN	AA-DQN, intensity	SPT-Threshold	SPT-Periodic	EDF-Periodic	EDF	Best learned − best rule
Nominal	–	13.04	**13.20**	12.17	12.61	12.30	12.25	11.41	4.24	+0.89
Attention depletion α	−30%	13.77	**13.91**	12.93	13.42	12.73	12.65	12.57	6.56	+1.18
	+30%	12.44	**12.50**	11.46	11.87	11.88	11.71	10.25	2.37	+0.61
Attention recovery β	−30%	**12.48**	12.47	11.60	11.93	11.82	11.60	10.05	4.24	+0.65
	+30%	13.50	**13.67**	12.57	13.07	12.59	12.67	12.44	4.24	+1.00
Fatigue accumulation δ	−30%	13.73	**13.85**	12.85	13.22	12.98	12.83	12.58	5.55	+0.86
	+30%	12.48	**12.55**	11.55	12.03	11.74	11.64	10.42	3.23	+0.81
Fatigue recovery η	−30%	12.61	**12.65**	11.80	12.21	11.92	11.80	10.68	4.24	+0.73
	+30%	13.38	**13.52**	12.44	12.87	12.58	12.55	12.10	4.24	+0.95
Fatigue penalty κ	−30%	13.81	**13.90**	12.97	13.41	12.81	12.64	12.49	5.16	+1.09
	+30%	12.39	**12.42**	11.28	11.72	11.80	11.71	10.40	3.32	+0.62
Circadian amplitude γ	−30%	13.24	**13.47**	12.33	12.82	12.43	12.41	11.81	4.49	+1.04
	+30%	12.85	**12.95**	12.01	12.39	12.14	11.98	11.01	3.99	+0.81
Fatigue coupling ϕ	−30%	13.12	**13.27**	12.28	12.71	12.35	12.30	11.56	4.57	+0.92
	+30%	12.97	**13.13**	12.08	12.52	12.25	12.18	11.27	3.89	+0.87

6.10. Computational cost

Training one AA-DQN agent for 5000 simulated workdays took 720 s on average on a single CPU thread (vanilla DQN: 614 s), including the periodic validation runs. AA-PPO needed only 171 s, about four times less, because it performs 80 minibatch gradient steps per 2048 decisions instead of one per decision. The complete evaluation loop, including the Python simulator, takes about 97 μs per decision when 500 days are evaluated in a batch. Even unbatched, inference is many orders of magnitude faster than the 10-minute decision interval, so the scheduler could run on a smartphone or wearable gateway. The heuristics are cheaper still, but their grid search (up to 252 parameter combinations $\times$ 300 days) must be repeated whenever the environment changes, just as the agent must be retrained.

7. Discussion

7.1. When and why does learning help?

The experiments support a more differentiated picture than "reinforcement learning beats heuristics". Four observations summarise it.

First, *the dominant effect of cognitive dynamics is that rest becomes a first-class scheduling decision.* Rules that ignore capacity lose a third to two thirds of the achievable return, and every competitive policy rests in 15–20% of slots. This conclusion is robust and arguably the most important for practice: any scheduler for human work should include a rest decision. Even a simple periodic or threshold rule recovers most of the value.

Second, *well-tuned, hand-designed rules are strong competitors*, and an evaluation that omits them overstates the value of learning. Our first internal experiments compared the agent against EDF-based rules only and suggested a clear advantage. That advantage shrank once SPT-based rest rules were added (Supplementary Section S7). The pre-specified hybrid AA-DQN ties with the best rule on return, and AA-PPO exceeds it by a moderate but highly consistent margin (all ten seeds, about two thirds of the test days). A naive memoryless DQN would have led to the opposite conclusion. The choice of learning algorithm thus matters as much as the choice of baselines: with the same observation, actions and reward, the policy-gradient learner outperformed the value-based one by almost one return unit, trained four times faster and generalised better to unseen workloads. These sensitivities echo concerns in the RL evaluation literature [60, 61] and in reviews of DRL scheduling [47, 49]. We recommend that studies of learned schedulers include tuned composite and state-aware rules as baselines.

Third, *what learning adds is the integration of several criteria that individ-*

ual rules trade off. SPT-based rules complete many tasks but starve demanding ones, whereas EDF-based rules protect demanding tasks but complete fewer. The learned policies achieve high count-based and difficulty-weighted on-time completion simultaneously (Fig. 5). Behaviourally, they do so by working intensively while the worker is fresh, scheduling demanding tasks when attention is high, filling low-attention periods with easy tasks, and concentrating rest around the circadian trough (Section 6.3). Each element is intuitive, but their state-, time- and deadline-dependent combination is hard to express as a rule, as the hand-crafted Cog-Heuristic illustrates.

Fourth, *the value of learning grows with the quality of the cognitive-state estimate.* With perfect observation, AA-DQN, which only ties with the best rule under nominal noise, outperforms it by more than one return unit while its own fatigue level drops from 0.33 to 0.27, whereas the rules gain nothing from better information. The observation history is the single most important architectural component, and longer histories or recurrent encoders help further. Partial observability, not the optimisation itself, is therefore the main bottleneck. This points to an engineering trade-off: investments in sensing and state estimation are what unlock the benefits of adaptive scheduling.

7.2. Implications for engineering practice

The results translate into design guidance for human-centric scheduling systems, such as knowledge-work assistants, workflow-management platforms and supervisory controllers in human–robot cells [27, 28, 136]. (i) *Always schedule rest.* A threshold or periodic rest rule on top of SPT-like sequencing is a strong, transparent default when cognitive-state estimates are noisy or unavailable. It requires only a small grid search to tune. (ii) *Use learning when state estimates*

are reliable or when several criteria must be balanced. Learned policies balance throughput, protection of demanding tasks and fatigue in ways that single rules cannot, and they profit from better sensors. Because results differ between learners, more than one algorithm should be tried; in this problem a policy-gradient learner with default settings was the better choice. (iii) *Make the well-being weight explicit.* The Pareto analysis shows that, up to a point, a higher weight on fatigue improves both well-being and on-time completion. The weight should be chosen deliberately with the stakeholders rather than left implicit in a heuristic's thresholds. (iv) *Evaluate fairly.* Common random numbers, disjoint tuning and test days, tuned rule baselines and multi-seed statistics are inexpensive and change conclusions.

7.3. Advantages and disadvantages of the approach

Table 10 summarises where the learned schedulers are, and are not, preferable to tuned rules. We discuss each point in turn.

Advantages. (i) *Performance.* The best learner exceeds the best of nine tuned rules on return, on-time rate and difficulty-weighted on-time rate (Section 6.2), with paired, multiplicity-corrected tests over 500 held-out days. The gain is moderate, not dramatic. Its source is the integration of several criteria that individual rules trade off: high throughput, protection of demanding tasks and well-timed rest. (ii) *No hand-design of the trade-off.* A rule needs a designer to decide how urgency, processing time, difficulty and the worker's state should be combined. The learner discovers a combination from the reward, and the weight λ makes the productivity–well-being trade-off explicit (Section 6.5). (iii) *Graceful degradation under sensor noise and parameter changes.* Because the policies integrate a history of observations, their performance hardly changes between perfect and

noisy state estimates, whereas reactive threshold rules lose up to one return unit (Section 6.8); AA-PPO also stays ahead of every rule when any cognitive parameter is changed by ±30% (Section 6.9). (iv) *Benefit from better sensing.* With accurate estimates, the advantage of learning grows markedly, whereas the rules gain nothing (Section 6.4). Investments in state estimation thus translate into better schedules only with an adaptive policy. (v) *Low run-time cost.* A decision takes microseconds and is independent of the number of tasks (Section 6.10).

Disadvantages. (i) *Generalisation to unseen workloads depends on the learner.* AA-DQN trained on 12 tasks per day loses to simple rules on 8 or 16 tasks, and domain randomisation did not close the gap. AA-PPO generalises much better but is still slightly behind the EDF-based rules on light days (Section 6.8). A tuned rule only needs to be re-tuned, which is cheap. (ii) *Training cost and model dependence.* Each learner needs about 2.4×10^5 simulated decisions and therefore a simulator that is faithful enough to the deployment setting, whereas a rule can be tuned on a few hundred days or even set by hand. (iii) *Transparency.* A threshold rule can be explained in one sentence and audited directly. The learned policies are neural networks, and their behaviour can only be described a posteriori (Section 6.3). (iv) *Seed variability and model selection.* Learned policies vary between training seeds and between checkpoints of one run, so validation-based selection and several seeds are needed (Section 6.2). (v) *Small margins in the nominal setting.* With realistic sensor noise, a well-tuned rest rule achieves most of the attainable return. When transparency or data are limiting, a tuned SPT rule with state-triggered rest is a strong and simple alternative.

In short, the approach is superior when cognitive-state estimates are reasonably accurate, when several criteria must be balanced, and when a simulator that

resembles the operating conditions is available. A tuned rest rule is preferable when conditions change frequently, when no faithful simulator is available, or when every decision must be explained.

Table 10: Advantages and disadvantages of the learned schedulers relative to tuned rule-based schedulers.

Aspect	Learned schedulers (AA-DQN, AA-PPO)	Tuned rules (e.g. SPT-Threshold)
Nominal performance	highest return and on-time rates (moderate margin)	close behind; most of the attainable return
Balancing criteria	combines throughput, hard-task protection and rest	each rule favours one criterion
Sensor noise	(nearly) insensitive (σ_o = 0–0.3)	threshold rules degrade with noise
Model parameters (±30%)	AA-PPO best in 14/14 perturbed models	ranking of rules changes
Better sensing	advantage grows markedly	no gain
Unseen workloads	AA-PPO near-best; AA-DQN degrades	robust after cheap re-tuning
Design effort	reward design, training, model selection	small grid search
Data requirement	faithful simulator, $\sim 2.4 \times 10^5$ decisions	a few hundred simulated days
Transparency	black box; explained a posteriori	fully transparent and auditable
Run-time cost	microseconds per decision	negligible

7.4. Ethical and organisational considerations

A scheduler that monitors attention and fatigue and recommends when a person should work on what and when to rest touches on autonomy, privacy and fairness, and these issues must be designed in rather than added later [121, 125, 147]. We see four requirements for responsible deployment. First, *the system should be advisory*. Recommendations should be presented to the worker, who retains the authority to accept or override them; evidence from human–robot teaming indicates that the allocation of decision authority affects both efficiency and satisfaction [134]. Second, *cognitive-state data are sensitive personal data*. Physiological and behavioural signals should be processed with informed consent, ideally on the worker's own device, and should never be used for performance appraisal. The observation model of the agents needs only two scalar estimates per slot, which supports such data minimisation. Third, *fairness across workers* requires that policies do not systematically disadvantage people whose fatigue dynamics differ from the nominal profile. Section 6.8 shows how performance changes for resilient and fatigue-prone profiles, and personalisation (Section 7.6) would further reduce such disparities. Fourth, *safety constraints* should bound the policy, for example by guaranteeing a minimum amount of rest per period regardless of deadline pressure, consistent with safe RL principles [148]. Notably, the learned policies in this study already increase rest relative to deadline-driven rules. The reward weight λ makes the productivity–well-being trade-off explicit and auditable (Section 6.5), so stakeholders can choose it deliberately rather than leaving it implicit in a heuristic.

7.5. Limitations

The study has limitations that define its scope. (i) *Simulation, not human data.* The cognitive model is grounded in established functional forms and its parameters

were chosen to be behaviourally plausible, but it has not been calibrated against individual time-series of attention and fatigue, and no established benchmark with such data exists. The absolute numbers should therefore be interpreted comparatively. The robustness and sensitivity analyses (Sections 6.8 and 6.9) show that the main conclusions hold under substantial changes of every cognitive parameter, but they cannot show that the model is correct. (ii) *A stylised cognitive model.* Attention and fatigue are each represented by one scalar with first-order dynamics. Effects such as motivation, task interest, learning, mind-wandering, sleep debt and inter-individual differences in the shape (not only the rate) of the dynamics are not modelled, and the circadian profile is the same for all workers. (iii) *Abstracted sensing.* Cognitive-state estimation is modelled as additive Gaussian noise on the true state. Real estimators can be biased, drift over the day, or fail intermittently, and they may be correlated with task type. (iv) *A single worker and a single day.* Multi-day carry-over of fatigue, sleep, and team-level interactions such as hand-offs and shared deadlines are outside the scope. (v) *Action abstraction.* The agent chooses a difficulty class or a dispatching rule rather than an individual task. This keeps the policy compact and size-invariant but excludes policies that, for example, deliberately abandon a specific hopeless task. (vi) *Algorithmic scope.* We study one value-based and one policy-gradient learner with a finite history; a recurrent encoder was tested only for AA-DQN. The PPO settings were not tuned, so the margin of AA-PPO may be a lower bound, while model-based or belief-tracking agents [65, 140] might extract more from the observation sequence. The environment ablations and most sensitivity studies were run with AA-DQN only. (vii) *No errors or interruptions.* Errors, rework and interruptions, which also shape real productivity [128, 129, 160], are not modelled. (viii) *Generalisation across*

workloads. AA-DQN policies trained on 12 tasks per day did not transfer well to 8 or 16 tasks, and simple domain randomisation did not resolve this; AA-PPO transferred much better, but we did not identify why, and the workloads tested were limited to 8–16 tasks (Section 6.8). (ix) *Post-hoc additions.* The intensity-only variant of AA-DQN was identified as promising in a component study before its full evaluation, and AA-PPO was added during the revision. Both were evaluated with the complete ten-seed protocol on the same held-out days, and AA-PPO was not tuned, but their results should still be confirmed on independent conditions.

7.6. Future work

Three extensions follow directly. First, *empirical calibration and validation*: fitting the latent dynamics to longitudinal data from vigilance probes, wearables and task logs [52, 55, 117], then evaluating the scheduler in a controlled user study with NASA-TLX and performance outcomes [56], ideally as a micro-randomised trial [130]. Second, *personalisation*: domain randomisation over worker profiles during training [161, 162], and Bayesian or meta-learning adaptation to an individual's parameters, in the spirit of personalised just-in-time interventions [131, 132]. Third, *multi-worker and human–robot settings*: extending the formulation to teams and to collaborative cells in which a robot absorbs demanding sub-tasks when a co-worker's attention is low [135–138], with constrained or multi-objective RL to enforce explicit well-being guarantees [63].

8. Conclusion

This paper formulated attention-aware task scheduling as a partially observable Markov decision process. In this formulation a worker's attention, fatigue and circadian state shape their capacity, rest is an explicit action, and the cognitive state

is observed only through noisy estimates. We proposed a memory-augmented deep reinforcement learning framework, instantiated it with a value-based learner (AA-DQN) and a policy-gradient learner (AA-PPO), and evaluated both against nine rule-based schedulers, including grid-tuned EDF and SPT rules with periodic or state-triggered rest. The evaluation used ten training seeds, 500 held-out workdays with common random numbers, reward-independent metrics and multiplicity-corrected tests.

Rules that ignore cognitive dynamics performed poorly, and every competitive policy rested in 15–20% of the day. Well-tuned rest rules were strong competitors: AA-DQN matched the return of the best of them and completed more tasks on time. AA-PPO exceeded the best rule in return (+0.73), on-time rate (75.1% versus 71.4%) and difficulty-weighted on-time rate (+7.0 percentage points), with every seed better than the rule. The learned policies are the only ones that combine high task throughput with protection of demanding tasks. They do hard work while the worker is fresh, fill the afternoon trough with easy tasks and rest before and during the post-lunch dip. AA-PPO remained the best method under sensor noise, for resilient and fatigue-prone workers, for heavier workloads and under $\pm 30\%$ changes of every cognitive parameter, whereas AA-DQN generalised poorly to unseen workloads. Controlled ablations showed that, within this simulator, modelling cognition is neither necessary nor sufficient for learning to help; the value of learning grows mainly with the reliability of the cognitive-state estimate, memory is the critical architectural component, and a moderate weight on fatigue improves both well-being and productivity.

The study is based on a literature-grounded but uncalibrated simulator. Its main limitation is therefore the gap to real workers, which future work should

close through calibration with physiological and behavioural data, personalised policies and user studies. The released simulator, agents and evaluation protocol provide a reproducible basis for that work and for fair comparisons of learned and hand-designed schedulers in human-centric systems.

CRediT authorship contribution statement

P. N. Bhargav Teja: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Lanka Sree Chathurya:** Methodology, Software, Validation, Data curation, Investigation, Writing – original draft. **Rajay Vedaraj I.S.:** Conceptualization, Supervision, Methodology, Resources, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethics statement

This study is entirely computational. It uses a simulated worker model whose parameters are taken from the published literature; no human participants, animals,

personal data or human-derived samples were involved, so ethical approval and informed consent were not required.

Data availability

The simulator, agents, baselines, experiment configurations, raw per-episode results and the scripts that generate every table and figure are openly available at `https://github.com/bhargavteja-9779/rl`. No human-subject data were collected.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to support code development, language editing and the organisation of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Acknowledgements

The authors thank Vellore Institute of Technology for providing the computational facilities used in this study.

References

[1] W. E. Smith, Various optimizers for single-stage production, Naval Research Logistics Quarterly 3 (1-2) (1956) 59–66. doi:10.1002/nav.3800030106.

[2] J. R. Jackson, Scheduling a production line to minimize maximum tardiness, Research Report 43, Management Science Research Project, University of California, Los Angeles (1955).

[3] R. Graham, E. Lawler, J. Lenstra, A. Kan, Optimization and Approximation in Deterministic Sequencing and Scheduling: a Survey, Elsevier, 1979, pp. 287–326. doi:10.1016/s0167-5060(08)70356-x.

[4] M. L. Pinedo, Scheduling, Springer International Publishing, 2016. doi:10.1007/978-3-319-26580-3.

[5] C. L. Liu, J. W. Layland, Scheduling algorithms for multiprogramming in a hard-real-time environment, Journal of the ACM 20 (1) (1973) 46–61. doi:10.1145/321738.321743.

[6] N. H. Mackworth, The breakdown of vigilance during prolonged visual search, Quarterly Journal of Experimental Psychology 1 (1) (1948) 6–21. doi:10.1080/17470214808416738.

[7] J. S. Warm, R. Parasuraman, G. Matthews, Vigilance requires hard mental work and is stressful, Human Factors: The Journal of the Human Factors and Ergonomics Society 50 (3) (2008) 433–441. doi:10.1518/001872008x312152.

[8] D. R. Thomson, D. Besner, D. Smilek, A resource-control account of sustained attention: Evidence from mind-wandering and vigilance paradigms, Perspectives on Psychological Science 10 (1) (2015) 82–96. doi:10.1177/1745691614556681.

[9] M. Esterman, D. Rothlein, Models of sustained attention, Current Opinion in Psychology 29 (2019) 174–180. doi:10.1016/j.copsyc.2019.03.005.

[10] M. A. Boksem, M. Tops, Mental fatigue: Costs and benefits, Brain Research Reviews 59 (1) (2008) 125–139. doi:10.1016/j.brainresrev.2008.07.001.

[11] M. M. Lorist, M. Klein, S. Nieuwenhuis, R. de Jong, G. Mulder, T. F. Meijman, Mental fatigue and task control: Planning and preparation, Psychophysiology 37 (5) (2000) 614–625. doi:10.1111/1469-8986.3750614.

[12] R. Hockey, The Psychology of Fatigue: Work, Effort and Control, Cambridge University Press, 2013. doi:10.1017/cbo9781139015394.

[13] P. L. Ackerman (Ed.), Cognitive Fatigue: Multidisciplinary Perspectives on Current Research and Future Applications, American Psychological Association, Washington, DC, 2011. doi:10.1037/12343-000.

[14] T. Åkerstedt, S. Folkard, The three-process model of alertness and its extension to performance, sleep latency, and sleep length, Chronobiology International 14 (2) (1997) 115–123. doi:10.3109/07420529709001149.

[15] T. H. Monk, The post-lunch dip in performance, Clinics in Sports Medicine 24 (2) (2005) e15–e23. doi:10.1016/j.csm.2004.12.002.

[16] C. Schmidt, F. Collette, C. Cajochen, P. Peigneux, A time to think: Circadian rhythms in human cognition, Cognitive Neuropsychology 24 (7) (2007) 755–789. doi:10.1080/02643290701754158.

[17] K. Blatter, C. Cajochen, Circadian rhythms in cognitive performance: Methodological constraints, protocols, theoretical underpinnings, Physiology & Behavior 90 (2-3) (2007) 196–208. doi:10.1016/j.physbeh.2006.09.009.

[18] R. A. HENNING, P. JACQUES, G. V. KISSEL, A. B. SULLIVAN, S. M. ALTERAS-WEBB, Frequent short rest breaks from computer work: effects on productivity and well-being at two field sites, Ergonomics 40 (1) (1997) 78–91. doi:10.1080/001401397188396.

[19] A. J. Dababneh, N. Swanson, R. L. Shell, Impact of added rest breaks on the productivity and well being of workers, Ergonomics 44 (2) (2001) 164–174. doi:10.1080/00140130121538.

[20] T. L. Galinsky, N. G. Swanson, S. L. Sauter, J. J. Hurrell, L. M. Schleifer, A field study of supplementary rest breaks for data-entry operators, Ergonomics 43 (5) (2000) 622–638. doi:10.1080/001401300184297.

[21] P. Tucker, The impact of rest breaks upon accident risk, fatigue and performance: A review, Work & Stress 17 (2) (2003) 123–137. doi:10.1080/0267837031000155949.

[22] P. Albulescu, I. Macsinga, A. Rusu, C. Sulea, A. Bodnaru, B. T. Tulbure, "give me a break!" a systematic review and meta-analysis on the efficacy of micro-breaks for increasing well-being and performance, PLOS ONE 17 (8) (2022) e0272460. doi:10.1371/journal.pone.0272460.

[23] D. Dawson, K. Reid, Fatigue, alcohol and performance impairment, Nature 388 (6639) (1997) 235–235. doi:10.1038/40775.

[24] A. Williamson, D. A. Lombardi, S. Folkard, J. Stutts, T. K. Courtney, J. L. Connor, The link between fatigue and safety, Accident Analysis & Prevention 43 (2) (2011) 498–515. doi:10.1016/j.aap.2009.11.011.

[25] M. Breque, L. De Nul, A. Petridis, Industry 5.0: Towards a sustainable, human-centric and resilient European industry, Tech. rep., European Commission, Directorate-General for Research and Innovation, Luxembourg (2021).

[26] X. Xu, Y. Lu, B. Vogel-Heuser, L. Wang, Industry 4.0 and industry 5.0—inception, conception and perception, Journal of Manufacturing Systems 61 (2021) 530–535. doi:10.1016/j.jmsy.2021.10.006.

[27] Y. Lu, H. Zheng, S. Chand, W. Xia, Z. Liu, X. Xu, L. Wang, Z. Qin, J. Bao, Outlook on human-centric manufacturing towards industry 5.0, Journal of Manufacturing Systems 62 (2022) 612–627. doi:10.1016/j.jmsy.2022.02.001.

[28] J. Leng, W. Sha, B. Wang, P. Zheng, C. Zhuang, Q. Liu, T. Wuest, D. Mourtzis, L. Wang, Industry 5.0: Prospect and retrospect, Journal of Manufacturing Systems 65 (2022) 279–295. doi:10.1016/j.jmsy.2022.09.017.

[29] D. Ivanov, The industry 5.0 framework: viability-based integration of the resilience, sustainability, and human-centricity perspectives, International Journal of Production Research 61 (5) (2022) 1683–1695. doi:10.1080/00207543.2022.2118892.

[30] M. Jaber, W. Neumann, Modelling worker fatigue and recovery in dual-resource constrained systems, Computers & Industrial Engineering 59 (1) (2010) 75–84. doi:10.1016/j.cie.2010.03.001.

[31] M. Jaber, Z. Givi, W. Neumann, Incorporating human fatigue and recovery into the learning–forgetting process, Applied Mathematical Modelling 37 (12-13) (2013) 7287–7299. doi:10.1016/j.apm.2013.02.028.

[32] Z. Givi, M. Jaber, W. Neumann, Modelling worker reliability with learning and fatigue, Applied Mathematical Modelling 39 (17) (2015) 5186–5199. doi:10.1016/j.apm.2015.03.038.

[33] S. E. Bechtold, R. E. Janaro, D. W. L. Sumners, Maximization of labor productivity through optimal rest-break schedules, Management Science 30 (12) (1984) 1442–1458. doi:10.1287/mnsc.30.12.1442.

[34] S. Konz, Work/rest: Part ii – the scientific basis (knowledge base) for the guide, International Journal of Industrial Ergonomics 22 (1-2) (1998) 73–99. doi:10.1016/s0169-8141(97)00069-3.

[35] M. Calzavara, A. Persona, F. Sgarbossa, V. Visentin, A model for rest allowance estimation to improve tasks assignment to operators, International Journal of Production Research 57 (3) (2018) 948–962. doi:10.1080/00207543.2018.1497816.

[36] E. J. Lodree, C. D. Geiger, X. Jiang, Taxonomy for integrating scheduling theory and human factors: Review and research opportunities, International Journal of Industrial Ergonomics 39 (1) (2009) 39–51. doi:10.1016/j.ergon.2008.05.001.

[37] W. Patrick Neumann, J. Dul, Human factors: spanning the gap between om and hrm, International Journal of Operations & Production Management 30 (9) (2010) 923–950. doi:10.1108/01443571011075056.

[38] J. Boudreau, W. Hopp, J. O. McClain, L. J. Thomas, On the interface between operations and human resources management, Manufacturing & Service Operations Management 5 (3) (2003) 179–202. doi:10.1287/msom.5.3.179.16032.

[39] N. Katiraee, M. Calzavara, S. Finco, D. Battini, O. Battaïa, Consideration of workers' differences in production systems modelling and design: State of the art and directions for future research, International Journal of Production Research 59 (11) (2021) 3237–3268. doi:10.1080/00207543.2021.1884766.

[40] H. Mao, M. Alizadeh, I. Menache, S. Kandula, Resource management with deep reinforcement learning, in: Proceedings of the 15th ACM Workshop on Hot Topics in Networks, HotNets-XV, ACM, 2016, pp. 50–56. doi:10.1145/3005745.3005750.

[41] H. Mao, M. Schwarzkopf, S. B. Venkatakrishnan, Z. Meng, M. Alizadeh, Learning scheduling algorithms for data processing clusters, in: Proceedings of the ACM Special Interest Group on Data Communication, SIGCOMM '19, ACM, 2019, pp. 270–288. doi:10.1145/3341302.3342080.

[42] B. Waschneck, A. Reichstaller, L. Belzner, T. Altenmüller, T. Bauernhansl, A. Knapp, A. Kyek, Optimization of global production scheduling with

deep reinforcement learning, Procedia CIRP 72 (2018) 1264–1269. doi:10.1016/j.procir.2018.03.212.

[43] S. Luo, Dynamic scheduling for flexible job shop with new job insertions by deep reinforcement learning, Applied Soft Computing 91 (2020) 106208. doi:10.1016/j.asoc.2020.106208.

[44] C. Zhang, W. Song, Z. Cao, J. Zhang, P. S. Tan, C. Xu, Learning to dispatch for job shop scheduling via deep reinforcement learning, in: Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 1621–1632.

[45] J. Park, J. Chun, S. H. Kim, Y. Kim, J. Park, Learning to schedule job-shop problems: representation and policy learning using graph neural network and reinforcement learning, International Journal of Production Research 59 (11) (2021) 3360–3377. doi:10.1080/00207543.2020.1870013.

[46] W. Song, X. Chen, Q. Li, Z. Cao, Flexible job-shop scheduling via graph neural network and deep reinforcement learning, IEEE Transactions on Industrial Informatics 19 (2) (2023) 1600–1610. doi:10.1109/tii.2022.3189725.

[47] B. M. Kayhan, G. Yildiz, Reinforcement learning applications to machine scheduling problems: a comprehensive literature review, Journal of Intelligent Manufacturing 34 (3) (2021) 905–929. doi:10.1007/s10845-021-01847-3.

[48] M. Panzer, B. Bender, Deep reinforcement learning in production systems: a systematic literature review, International Journal of Production Research 60 (13) (2021) 4316–4341. doi:10.1080/00207543.2021.1973138.

[49] M. Khadivi, T. Charter, M. Yaghoubi, M. Jalayer, M. Ahang, A. Shojaeinasab, H. Najjaran, Deep reinforcement learning for machine scheduling: Methodology, the state-of-the-art, and future directions, Computers & Industrial Engineering 200 (2025) 110856. doi:10.1016/j.cie.2025.110856.

[50] W. Tan, X. Yuan, J. Wang, X. Zhang, A fatigue-conscious dual resource constrained flexible job shop scheduling problem by enhanced nsga-ii: An application from casting workshop, Computers & Industrial Engineering 160 (2021) 107557. doi:10.1016/j.cie.2021.107557.

[51] Y. Hu, Z. Zhang, J. Chen, D. Tang, Q. Cai, Graph-based deep reinforcement learning for dynamic scheduling of flexible job-shop considering worker fatigue and multi-skill factors, Applied Soft Computing 184 (2025) 113712. doi:10.1016/j.asoc.2025.113712.

[52] R. L. Charles, J. Nixon, Measuring mental workload using physiological measures: A systematic review, Applied Ergonomics 74 (2019) 221–232. doi:10.1016/j.apergo.2018.08.028.

[53] G. Borghini, L. Astolfi, G. Vecchiato, D. Mattia, F. Babiloni, Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness, Neuroscience & Biobehavioral Reviews 44 (2014) 58–75. doi:10.1016/j.neubiorev.2012.10.003.

[54] M. A. Hogervorst, A.-M. Brouwer, J. B. F. van Erp, Combining and comparing eeg, peripheral physiology and eye-related measures for the as-

sessment of mental workload, Frontiers in Neuroscience 8 (Oct. 2014). doi:10.3389/fnins.2014.00322.

[55] M. Basner, D. F. Dinges, Maximizing sensitivity of the psychomotor vigilance test (pvt) to sleep loss, Sleep 34 (5) (2011) 581–591. doi:10.1093/sleep/34.5.581.

[56] S. G. Hart, L. E. Staveland, Development of NASA-TLX (task load index): Results of empirical and theoretical research, in: P. A. Hancock, N. Meshkati (Eds.), Human Mental Workload, Vol. 52 of Advances in Psychology, North-Holland, Amsterdam, 1988, pp. 139–183. doi:10.1016/s0166-4115(08)62386-9.

[57] K. Åström, Optimal control of markov processes with incomplete state information, Journal of Mathematical Analysis and Applications 10 (1) (1965) 174–205. doi:10.1016/0022-247x(65)90154-x.

[58] R. D. Smallwood, E. J. Sondik, The optimal control of partially observable markov processes over a finite horizon, Operations Research 21 (5) (1973) 1071–1088. doi:10.1287/opre.21.5.1071.

[59] L. P. Kaelbling, M. L. Littman, A. R. Cassandra, Planning and acting in partially observable stochastic domains, Artificial Intelligence 101 (1-2) (1998) 99–134. doi:10.1016/s0004-3702(98)00023-x.

[60] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, D. Meger, Deep reinforcement learning that matters, Proceedings of the AAAI Conference on Artificial Intelligence 32 (1) (Apr. 2018). doi:10.1609/aaai.v32i1.11694.

[61] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, M. G. Bellemare, Deep reinforcement learning at the edge of the statistical precipice, in: Advances in Neural Information Processing Systems, Vol. 34, 2021, pp. 29304–29320.

[62] D. M. Roijers, P. Vamplew, S. Whiteson, R. Dazeley, A survey of multi-objective sequential decision-making, Journal of Artificial Intelligence Research 48 (2013) 67–113. doi:10.1613/jair.3987.

[63] C. F. Hayes, R. Rădulescu, E. Bargiacchi, J. Källström, M. Macfarlane, M. Reymond, T. Verstraeten, L. M. Zintgraf, R. Dazeley, F. Heintz, E. Howley, A. A. Irissappane, P. Mannion, A. Nowé, G. Ramos, M. Restelli, P. Vamplew, D. M. Roijers, A practical guide to multi-objective reinforcement learning and planning, Autonomous Agents and Multi-Agent Systems 36 (1) (Apr. 2022). doi:10.1007/s10458-022-09552-y.

[64] S. P. Singh, T. Jaakkola, M. I. Jordan, Learning Without State-Estimation in Partially Observable Markovian Decision Processes, Elsevier, 1994, pp. 284–292. doi:10.1016/b978-1-55860-335-6.50042-8.

[65] M. Hausknecht, P. Stone, Deep recurrent Q-learning for partially observable MDPs, in: AAAI Fall Symposium on Sequential Decision Making for Intelligent Agents, 2015, pp. 29–37.

[66] T. Ni, B. Eysenbach, R. Salakhutdinov, Recurrent model-free RL can be a strong baseline for many POMDPs, in: Proceedings of the 39th International Conference on Machine Learning, Vol. 162 of PMLR, 2022, pp. 16691–16723.

[67] H. Van Hasselt, A. Guez, D. Silver, Deep reinforcement learning with double q-learning, Proceedings of the AAAI Conference on Artificial Intelligence 30 (1) (Mar. 2016). doi:10.1609/aaai.v30i1.10295.

[68] Z. Wang, T. Schaul, M. Hessel, H. van Hasselt, M. Lanctot, N. de Freitas, Dueling network architectures for deep reinforcement learning, in: Proceedings of the 33rd International Conference on Machine Learning, Vol. 48 of PMLR, 2016, pp. 1995–2003.

[69] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, arXiv preprint arXiv:1707.06347 (2017).

[70] G. Zhang, D. Zhang, Z. Li, N. Yu, W. Zhang, An evolutionary algorithm with neighborhood structure incorporating reinforcement learning for dual resource constrained flexible job shop scheduling problem with worker transportation fatigue, Expert Systems with Applications 301 (2026) 130523. doi:10.1016/j.eswa.2025.130523.

[71] Z. Shao, Z. Zhang, W. Shao, Fatigue and energy conscious multi-objective hybrid flow-shop scheduling: an offline deep-q-network-based metaheuristic, Engineering Optimization (2026) 1–37doi:10.1080/0305215x.2026.2693129.

[72] K.-T. Park, C. Lim, J.-Y. Lee, Production scheduling for human–robot collaborative assembly workstations under constraints of ergonomic fatigue and simultaneous cooperation, Journal of Manufacturing Systems 83 (2025) 337–356. doi:10.1016/j.jmsy.2025.09.012.

[73] J. M. Moore, An n job, one machine sequencing algorithm for minimizing the number of late jobs, Management Science 15 (1) (1968) 102–109. doi:10.1287/mnsc.15.1.102.

[74] G. C. Buttazzo, Hard Real-Time Computing Systems: Predictable Scheduling Algorithms and Applications, Springer US, 2011. doi:10.1007/978-1-4614-0676-1.

[75] S. S. Panwalkar, W. Iskander, A survey of scheduling rules, Operations Research 25 (1) (1977) 45–61. doi:10.1287/opre.25.1.45.

[76] R. Haupt, A survey of priority rule-based scheduling, OR Spektrum 11 (1) (1989) 3–16. doi:10.1007/bf01721162.

[77] G. E. Vieira, J. W. Herrmann, E. Lin, Rescheduling manufacturing systems: A framework of strategies, policies, and methods, Journal of Scheduling 6 (1) (2003) 39–62. doi:10.1023/a:1022235519958.

[78] D. Ouelhadj, S. Petrovic, A survey of dynamic scheduling in manufacturing systems, Journal of Scheduling 12 (4) (2008) 417–431. doi:10.1007/s10951-008-0090-8.

[79] A. Ernst, H. Jiang, M. Krishnamoorthy, D. Sier, Staff scheduling and rostering: A review of applications, methods and models, European Journal of Operational Research 153 (1) (2004) 3–27. doi:10.1016/s0377-2217(03)00095-x.

[80] E. K. Burke, P. De Causmaecker, G. V. Berghe, H. Van Landeghem, The state of the art of nurse rostering, Journal of Scheduling 7 (6) (2004) 441–499. doi:10.1023/b:josh.0000046076.75950.0b.

[81] J. Van den Bergh, J. Beliën, P. De Bruecker, E. Demeulemeester, L. De Boeck, Personnel scheduling: A literature review, European Journal of Operational Research 226 (3) (2013) 367–385. doi:10.1016/j.ejor.2012.11.029.

[82] T. P. WRIGHT, Factors affecting the cost of airplanes, Journal of the Aeronautical Sciences 3 (4) (1936) 122–128. doi:10.2514/8.155.

[83] T. Ren, Z. Dong, F. Qi, J. Weng, H. Xue, Solving the flow-shop scheduling problem with human factors and two competing agents with deep reinforcement learning, Engineering Optimization 57 (3) (2024) 671–687. doi:10.1080/0305215x.2024.2329998.

[84] Z. Yi, K. Deng, Y. Lei, J. Zhang, L. Lu, Q. Deng, An end-to-end deep reinforcement learning method for dynamic flexible job shop scheduling problem considering multi-skilled worker constraints, Journal of Intelligent Manufacturing (Mar. 2026). doi:10.1007/s10845-025-02784-1.

[85] C. Ding, F. Qiao, D. Wang, J. Liu, A novel hybrid intelligent scheduling: integrating human feedback into reinforcement learning for adaptive preference objectives, International Journal of Production Research 63 (16) (2025) 6037–6055. doi:10.1080/00207543.2025.2467448.

[86] X. Chang, X. Jia, J. Ren, A reinforcement learning enhanced memetic algorithm for multi-objective flexible job shop scheduling toward industry 5.0, International Journal of Production Research 63 (1) (2024) 119–147. doi:10.1080/00207543.2024.2357740.

[87] J. Wang, Y. Yan, Y. Hu, X. Yang, L. Zhang, A transfer reinforcement learning and digital-twin based task allocation method for human-robot collaboration assembly, Engineering Applications of Artificial Intelligence 144 (2025) 110064. doi:10.1016/j.engappai.2025.110064.

[88] J. Wang, Y. Yan, Y. Hu, X. Yang, A reinforcement learning from human feedback based method for task allocation of human robot collaboration assembly considering human preference, Advanced Engineering Informatics 66 (2025) 103497. doi:10.1016/j.aei.2025.103497.

[89] R. Wang, D. Zhao, A. Gupte, B.-C. Min, Initial task allocation in multi-human multi-robot teams: An attention-enhanced hierarchical reinforcement learning approach, IEEE Robotics and Automation Letters 9 (4) (2024) 3451–3458. doi:10.1109/lra.2024.3366414.

[90] J. Xue, X. Li, N. Zhang, Safe reinforcement learning with online filtering for fatigue-predictive human–robot task planning and allocation in production, Journal of Manufacturing Systems 84 (2026) 561–583. doi:10.1016/j.jmsy.2025.12.019.

[91] E. H. Grosse, C. H. Glock, W. P. Neumann, Human factors in order picking: a content analysis of the literature, International Journal of Production Research 55 (5) (2016) 1260–1276. doi:10.1080/00207543.2016.1186296.

[92] B. A. Kadir, O. Broberg, C. S. d. Conceição, Current research and future perspectives on human factors and ergonomics in industry 4.0, Computers &

Industrial Engineering 137 (2019) 106004. doi:10.1016/j.cie.2019.106004.

[93] W. P. Neumann, S. Winkelhaus, E. H. Grosse, C. H. Glock, Industry 4.0 and the human factor – a systems framework and analysis methodology for successful development, International Journal of Production Economics 233 (2021) 107992. doi:10.1016/j.ijpe.2020.107992.

[94] W. Zhang, T. G. Dietterich, A reinforcement learning approach to job-shop scheduling, in: Proceedings of the 14th International Joint Conference on Artificial Intelligence, 1995, pp. 1114–1120.

[95] M. Aydin, E. Öztemel, Dynamic job-shop scheduling using reinforcement learning agents, Robotics and Autonomous Systems 33 (2-3) (2000) 169–178. doi:10.1016/s0921-8890(00)00087-7.

[96] C. J. C. H. Watkins, P. Dayan, Q-learning, Machine Learning 8 (3-4) (1992) 279–292. doi:10.1007/bf00992698.

[97] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, D. Hassabis, Human-level control through deep reinforcement learning, Nature 518 (7540) (2015) 529–533. doi:10.1038/nature14236.

[98] C.-L. Liu, C.-C. Chang, C.-J. Tseng, Actor-critic deep reinforcement learning for solving job shop scheduling problems, IEEE Access 8 (2020) 71752–71762. doi:10.1109/access.2020.2987820.

[99] J. Shahrabi, M. A. Adibi, M. Mahootchi, A reinforcement learning approach to parameter estimation in dynamic job shop scheduling, Computers & Industrial Engineering 110 (2017) 75–82. doi:10.1016/j.cie.2017.05.026.

[100] Y. Bengio, A. Lodi, A. Prouvost, Machine learning for combinatorial optimization: A methodological tour d'horizon, European Journal of Operational Research 290 (2) (2021) 405–421. doi:10.1016/j.ejor.2020.07.063.

[101] N. Mazyavkina, S. Sviridov, S. Ivanov, E. Burnaev, Reinforcement learning for combinatorial optimization: A survey, Computers & Operations Research 134 (2021) 105400. doi:10.1016/j.cor.2021.105400.

[102] J. v. Remmerden, Z. Bukhsh, Y. Zhang, Offline reinforcement learning for learning to dispatch for job shop scheduling, Machine Learning 114 (8) (2025). doi:10.1007/s10994-025-06826-w.

[103] D. Kahneman, Attention and Effort, Prentice-Hall, Englewood Cliffs, NJ, 1973.

[104] C. D. Wickens, Multiple resources and mental workload, Human Factors: The Journal of the Human Factors and Ergonomics Society 50 (3) (2008) 449–455. doi:10.1518/001872008x288394.

[105] J. Sweller, Cognitive load during problem solving: Effects on learning, Cognitive Science 12 (2) (1988) 257–285. doi:10.1207/s15516709cog1202_4.

[106] A. Ariga, A. Lleras, Brief and rare mental "breaks" keep you focused: Deactivation and reactivation of task goals preempt vigilance decrements, Cognition 118 (3) (2011) 439–443. doi:10.1016/j.cognition.2010.12.007.

[107] R. Kurzban, A. Duckworth, J. W. Kable, J. Myers, An opportunity cost model of subjective effort and task performance, Behavioral and Brain Sciences 36 (6) (2013) 661–679. doi:10.1017/s0140525x12003196.

[108] J. F. Hopstaken, D. van der Linden, A. B. Bakker, M. A. J. Kompier, A multifaceted investigation of the link between mental fatigue and task disengagement, Psychophysiology 52 (3) (2014) 305–315. doi:10.1111/psyp.12339.

[109] A. A. Borbély, A two process model of sleep regulation, Human Neurobiology 1 (3) (1982) 195–204.

[110] S. Daan, D. G. Beersma, A. A. Borbely, Timing of human sleep: recovery process gated by a circadian pacemaker, American Journal of Physiology-Regulatory, Integrative and Comparative Physiology 246 (2) (1984) R161–R183. doi:10.1152/ajpregu.1984.246.2.r161.

[111] A. A. Borbély, S. Daan, A. Wirz-Justice, T. Deboer, The two-process model of sleep regulation: a reappraisal, Journal of Sleep Research 25 (2) (2016) 131–143. doi:10.1111/jsr.12371.

[112] S. R. Hursh, D. P. Redmond, M. L. Johnson, D. R. Thorne, G. Belenky, T. J. Balkin, W. F. Storm, J. C. Miller, D. R. Eddy, Fatigue models for applied re-

search in warfighting, Aviation, Space, and Environmental Medicine 75 (3, Suppl.) (2004) A44–A53.

[113] S. Sonnentag, C. Fritz, The recovery experience questionnaire: Development and validation of a measure for assessing recuperation and unwinding from work., Journal of Occupational Health Psychology 12 (3) (2007) 204–221. doi:10.1037/1076-8998.12.3.204.

[114] C. Fritz, A. M. Ellis, C. A. Demsky, B. C. Lin, F. Guros, Embracing work breaks, Organizational Dynamics 42 (4) (2013) 274–280. doi:10.1016/j.orgdyn.2013.07.005.

[115] S. Kim, Y. Park, Q. Niu, Micro-break activities at work to recover from daily work demands, Journal of Organizational Behavior 38 (1) (2016) 28–44. doi:10.1002/job.2109.

[116] A. Gevins, M. E. Smith, Neurophysiological measures of cognitive workload during human-computer interaction, Theoretical Issues in Ergonomics Science 4 (1-2) (2003) 113–131. doi:10.1080/14639220210159717.

[117] F. Shaffer, J. P. Ginsberg, An overview of heart rate variability metrics and norms, Frontiers in Public Health 5 (2017). doi:10.3389/fpubh.2017.00258.

[118] D. F. Dinges, J. W. Powell, Microcomputer analyses of performance on a portable, simple visual rt task during sustained operations, Behavior Research Methods, Instruments, & Computers 17 (6) (1985) 652–655. doi:10.3758/bf03200977.

[119] M. S. Young, K. A. Brookhuis, C. D. Wickens, P. A. Hancock, State of science: mental workload in ergonomics, Ergonomics 58 (1) (2014) 1–17. doi:10.1080/00140139.2014.956151.

[120] E. A. Byrne, R. Parasuraman, Psychophysiology and adaptive automation, Biological Psychology 42 (3) (1996) 249–268. doi:10.1016/0301-0511(95)05161-9.

[121] R. Parasuraman, T. Sheridan, C. Wickens, A model for types and levels of human interaction with automation, IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans 30 (3) (2000) 286–297. doi:10.1109/3468.844354.

[122] D. B. Kaber, M. R. Endsley, The effects of level of automation and adaptive automation on human performance, situation awareness and workload in a dynamic control task, Theoretical Issues in Ergonomics Science 5 (2) (2004) 113–153. doi:10.1080/1463922021000054335.

[123] S. H. Fairclough, Fundamentals of physiological computing, Interacting with Computers 21 (1-2) (2009) 133–145. doi:10.1016/j.intcom.2008.10.011.

[124] T. O. Zander, C. Kothe, Towards passive brain–computer interfaces: applying brain–computer interface technology to human–machine systems in general, Journal of Neural Engineering 8 (2) (2011) 025005. doi:10.1088/1741-2560/8/2/025005.

[125] K. M. Feigh, M. C. Dorneich, C. C. Hayes, Toward a characterization of adaptive systems: A framework for researchers and system designers,

Human Factors: The Journal of the Human Factors and Ergonomics Society 54 (6) (2012) 1008–1024. doi:10.1177/0018720812443983.

[126] E. Horvitz, J. Apacible, Learning and reasoning about interruption, in: Proceedings of the 5th international conference on Multimodal interfaces, ICMI-PUI03, ACM, 2003, pp. 20–27. doi:10.1145/958432.958440.

[127] S. T. Iqbal, B. P. Bailey, Effects of intelligent notification management on users and their tasks, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, CHI '08, ACM, 2008, pp. 93–102. doi:10.1145/1357054.1357070.

[128] M. Czerwinski, E. Horvitz, S. Wilhite, A diary study of task switching and interruptions, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, CHI04, ACM, 2004, pp. 175–182. doi:10.1145/985692.985715.

[129] G. Mark, D. Gudith, U. Klocke, The cost of interrupted work: more speed and stress, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, CHI '08, ACM, 2008, pp. 107–110. doi:10.1145/1357054.1357072.

[130] P. Klasnja, E. B. Hekler, S. Shiffman, A. Boruvka, D. Almirall, A. Tewari, S. A. Murphy, Microrandomized trials: An experimental design for developing just-in-time adaptive interventions., Health Psychology 34 (Suppl) (2015) 1220–1228. doi:10.1037/hea0000305.

[131] I. Nahum-Shani, S. N. Smith, B. J. Spring, L. M. Collins, K. Witkiewitz, A. Tewari, S. A. Murphy, Just-in-time adaptive interventions (jitais) in

mobile health: Key components and design principles for ongoing health behavior support, Annals of Behavioral Medicine 52 (6) (2017) 446–462. doi:10.1007/s12160-016-9830-8.

[132] P. Liao, K. Greenewald, P. Klasnja, S. Murphy, Personalized heartsteps: A reinforcement learning algorithm for optimizing physical activity, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 4 (1) (2020) 1–22. doi:10.1145/3381007.

[133] C. Yu, J. Liu, S. Nemati, G. Yin, Reinforcement learning in healthcare: A survey, ACM Computing Surveys 55 (1) (2021) 1–36. doi:10.1145/3477600.

[134] M. C. Gombolay, R. A. Gutierrez, S. G. Clarke, G. F. Sturla, J. A. Shah, Decision-making authority, team efficiency and human worker satisfaction in mixed human–robot teams, Autonomous Robots 39 (3) (2015) 293–312. doi:10.1007/s10514-015-9457-9.

[135] L. Peternel, N. Tsagarakis, D. Caldwell, A. Ajoudani, Robot adaptation to human physical fatigue in human–robot co-manipulation, Autonomous Robots 42 (5) (2017) 1011–1021. doi:10.1007/s10514-017-9678-1.

[136] A. Ajoudani, A. M. Zanchettin, S. Ivaldi, A. Albu-Schäffer, K. Kosuge, O. Khatib, Progress and prospects of the human–robot collaboration, Autonomous Robots 42 (5) (2017) 957–975. doi:10.1007/s10514-017-9677-2.

[137] E. Lamon, A. De Franco, L. Peternel, A. Ajoudani, A capability-aware role allocation approach to industrial assembly tasks, IEEE Robotics and

Automation Letters 4 (4) (2019) 3378–3385. doi:10.1109/lra.2019.2926963.

[138] R. Zhang, Q. Lv, J. Li, J. Bao, T. Liu, S. Liu, A reinforcement learning method for human-robot collaboration in assembly tasks, Robotics and Computer-Integrated Manufacturing 73 (2022) 102227. doi:10.1016/j.rcim.2021.102227.

[139] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Computation 9 (8) (1997) 1735–1780. doi:10.1162/neco.1997.9.8.1735.

[140] M. Igl, L. Zintgraf, T. A. Le, F. Wood, S. Whiteson, Deep variational reinforcement learning for POMDPs, in: Proceedings of the 35th International Conference on Machine Learning, Vol. 80 of PMLR, 2018, pp. 2117–2126.

[141] L.-J. Lin, Self-improving reactive agents based on reinforcement learning, planning and teaching, Machine Learning 8 (3-4) (1992) 293–321. doi:10.1007/bf00992699.

[142] M. Hessel, J. Modayil, H. Van Hasselt, T. Schaul, G. Ostrovski, W. Dabney, D. Horgan, B. Piot, M. Azar, D. Silver, Rainbow: Combining improvements in deep reinforcement learning, Proceedings of the AAAI Conference on Artificial Intelligence 32 (1) (Apr. 2018). doi:10.1609/aaai.v32i1.11796.

[143] J. Tsitsiklis, B. Van Roy, An analysis of temporal-difference learning with function approximation, IEEE Transactions on Automatic Control 42 (5) (1997) 674–690. doi:10.1109/9.580874.

[144] F. Wu, S. Li, Y. Zhang, J. Tang, P. Yuan, Y. Hu, L. Wu, Heterogeneous operator mechanism and stochastic configuration network-enhanced large-scale multi-objective evolutionary algorithm, Journal of Computational Design and Engineering 13 (2) (2025) 1–44. doi:10.1093/jcde/qwaf144.

[145] P. Vamplew, R. Dazeley, A. Berry, R. Issabekov, E. Dekker, Empirical evaluation methods for multiobjective reinforcement learning algorithms, Machine Learning 84 (1-2) (2010) 51–80. doi:10.1007/s10994-010-5232-5.

[146] A. Y. Ng, D. Harada, S. Russell, Policy invariance under reward transformations: Theory and application to reward shaping, in: Proceedings of the 16th International Conference on Machine Learning, 1999, pp. 278–287.

[147] D. Amodei, C. Olah, J. Steinhardt, P. Christiano, J. Schulman, D. Mané, Concrete problems in AI safety, arXiv preprint arXiv:1606.06565 (2016).

[148] J. García, F. Fernández, A comprehensive survey on safe reinforcement learning, Journal of Machine Learning Research 16 (42) (2015) 1437–1480.

[149] G. Dulac-Arnold, N. Levine, D. J. Mankowitz, J. Li, C. Paduraru, S. Gowal, T. Hester, Challenges of real-world reinforcement learning: definitions, benchmarks and analysis, Machine Learning 110 (9) (2021) 2419–2468. doi:10.1007/s10994-021-05961-4.

[150] H. P. Van Dongen, G. Maislin, J. M. Mullington, D. F. Dinges, The cumulative cost of additional wakefulness: Dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restric-

tion and total sleep deprivation, Sleep 26 (2) (2003) 117–126. doi:10.1093/sleep/26.2.117.

[151] M. L. Puterman, Markov Decision Processes: Discrete Stochastic Dynamic Programming, Wiley, 1994. doi:10.1002/9780470316887.

[152] R. S. Sutton, A. G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, Cambridge, MA, 2018.

[153] P. J. Huber, Robust estimation of a location parameter, The Annals of Mathematical Statistics 35 (1) (1964) 73–101. doi:10.1214/aoms/1177703732.

[154] D. P. Kingma, J. Ba, Adam: A method for stochastic optimization, in: International Conference on Learning Representations, 2015.

[155] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, P. Abbeel, High-dimensional continuous control using generalized advantage estimation, in: Proceedings of the 4th International Conference on Learning Representations (ICLR), 2016.

[156] B. Efron, Bootstrap methods: Another look at the jackknife, The Annals of Statistics 7 (1) (Jan. 1979). doi:10.1214/aos/1176344552.

[157] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (6) (1945) 80. doi:10.2307/3001968.

[158] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.

[159] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, et al., PyTorch: An imperative style, high-performance deep learning library, in: Advances in Neural Information Processing Systems, Vol. 32, 2019, pp. 8024–8035.

[160] V. M. González, G. Mark, "constant, constant, multi-tasking craziness": managing multiple working spheres, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, CHI04, ACM, 2004, pp. 113–120. doi:10.1145/985692.985707.

[161] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, P. Abbeel, Domain randomization for transferring deep neural networks from simulation to the real world, in: 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2017, pp. 23–30. doi:10.1109/iros.2017.8202133.

[162] W. Zhao, J. P. Queralta, T. Westerlund, Sim-to-real transfer in deep reinforcement learning for robotics: a survey, in: 2020 IEEE Symposium Series on Computational Intelligence (SSCI), IEEE, 2020, pp. 737–744. doi:10.1109/ssci47803.2020.9308468.